

statistics and probability

Presented By
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➔ why math is very important ?

- 1- Statistics & Probability are the foundation of Machine Learning
 - 2- ML models build on Statistics & Probability
 - 3- Pre-steps before ML models also depend on Statistics & Probability
(mean , median , std, distrubition , outliers)
 - 4- Feature Engineering aslo need Statistics & Probability
 - 5- Understanding Overfitting using math.
 - 6- Loss Functions Come From Statistics.
 - 7- After training a model, we need to know:
 - Is the result really good?
 - Or is it just random?
-

➔ What is Statistics?

Statistics = collecting, organizing, analyzing, and understanding data.

It helps answer:

- What is the average salary?
- Are sales increasing?



- **Statistics: It is the science by which the past is analyzed**
-

➔ What is Probability?

Probability = the chance that something happens.

It helps answer:

- What is the chance a customer buys a product?
- What is the probability an email is spam?
- What is the chance it rains tomorrow?



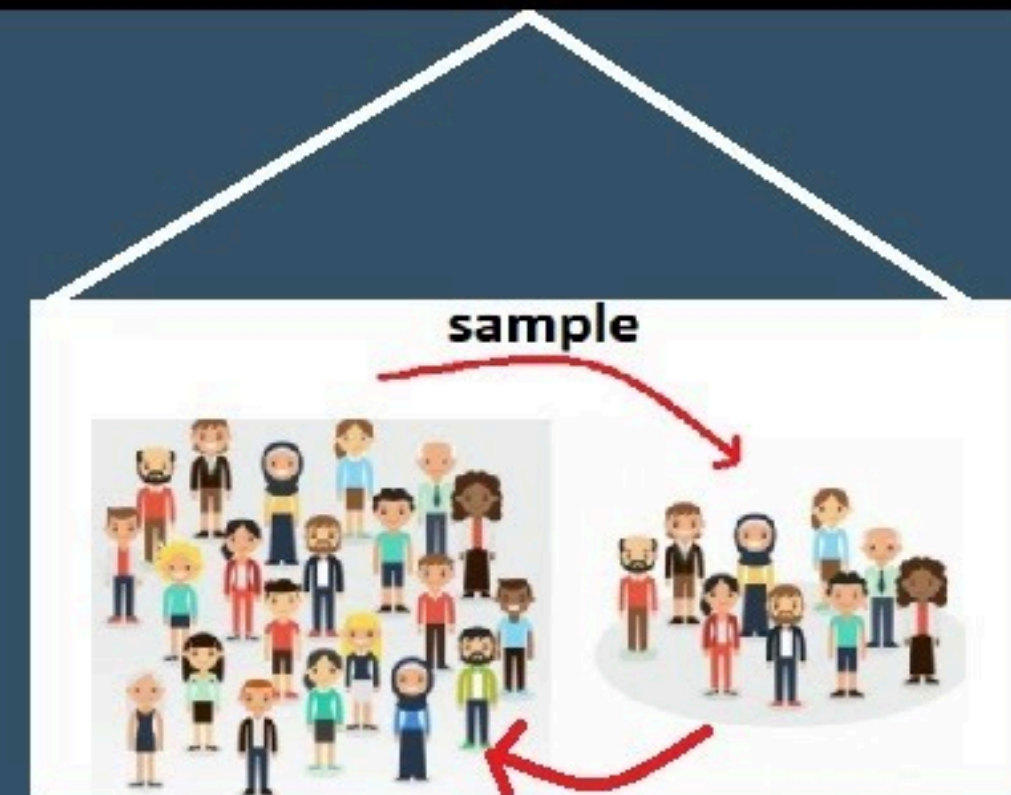
- **Probability: It is the other side of statistics because it predicts the future.**

Descriptive Statistics



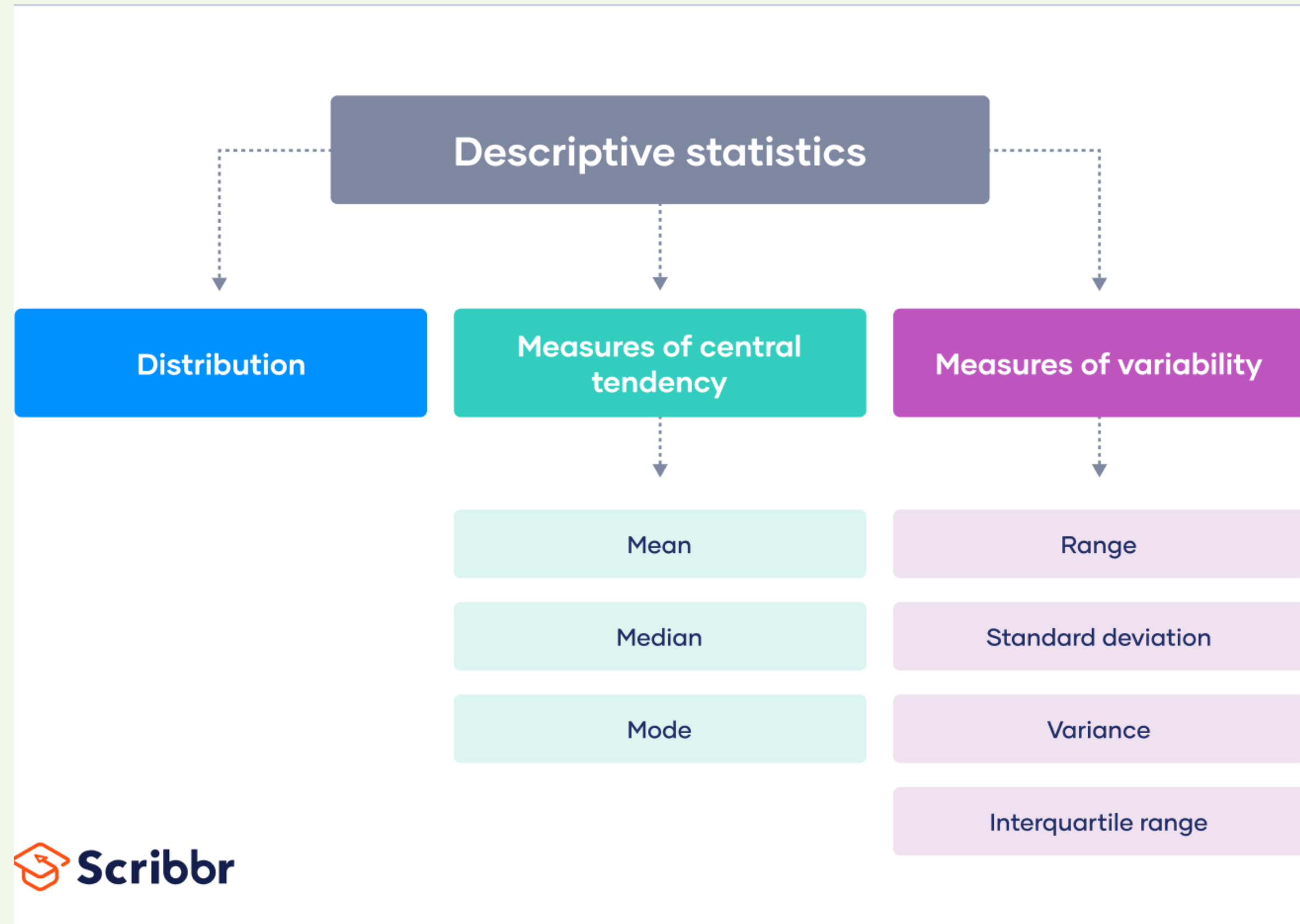
VS

Inferential Statistics



Descriptive Statistics

- Descriptive Statistics is a branch of Statistics used to summarize, organize, and describe data.
- Instead of looking at thousands of rows manually, descriptive statistics gives us simple numbers and charts to understand the data quickly.



Population:

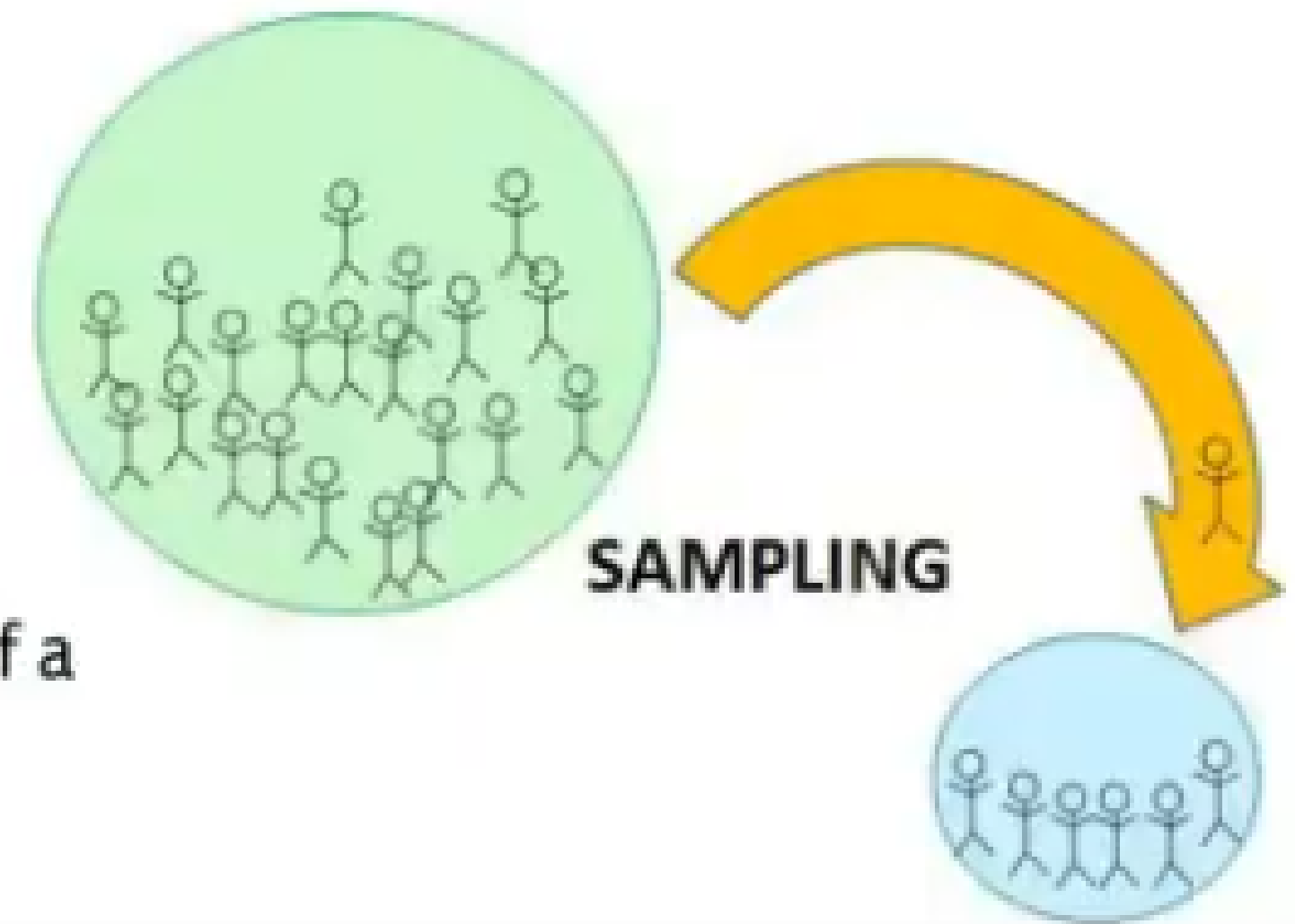
The entire group of individuals or items being studied.

Example: All customers of a bank.

Sample:

A subset of the population selected for analysis.

Example: 1,000 customers selected from all customers of a bank.



Measure of Central Tendency



→ Mean

Population Mean	Sample Mean
$\mu = \frac{\sum_{i=1}^N x_i}{N}$ N = number of items in the population	$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$ n = number of items in the sample

Measure of Central Tendency



→ Median

Median: The middle value when data is ordered.

- If the dataset size is odd, the median is the middle value.
- If even, it is the average of the two middle values.

1, 3, 3, **6**, 7, 8, 9

Median = **6**

1, 2, 3, **4**, **5**, 6, 8, 9

Median = $(4 + 5) \div 2$

= **4.5**

Measure of Central Tendency



Mode

Mode: The value(s) that appear most frequently in the dataset.

Example: The most common shoe size in a store.

1, 3, 3, 3, 5, 6, 6, 9, 9, 9

There are two modes

3 9

Measure Of Variability



➔ Range



Measure Of Variability



➔ Variance

- How far the data values are spread out from the mean (average)

Variance	Meaning
Small Variance	Data points close to mean
Large Variance	Data points far from mean

Variance: The average of squared differences from the mean.

Population

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}$$

Sample

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}$$



Measure Of Variability



→ Variance

Symbol	Meaning
x	Data value
μ	Mean
N	Number of values
$(x-\mu)$	Distance from mean

Variance: The average of squared differences from the mean.

Population

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}$$

Sample

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N-1}$$



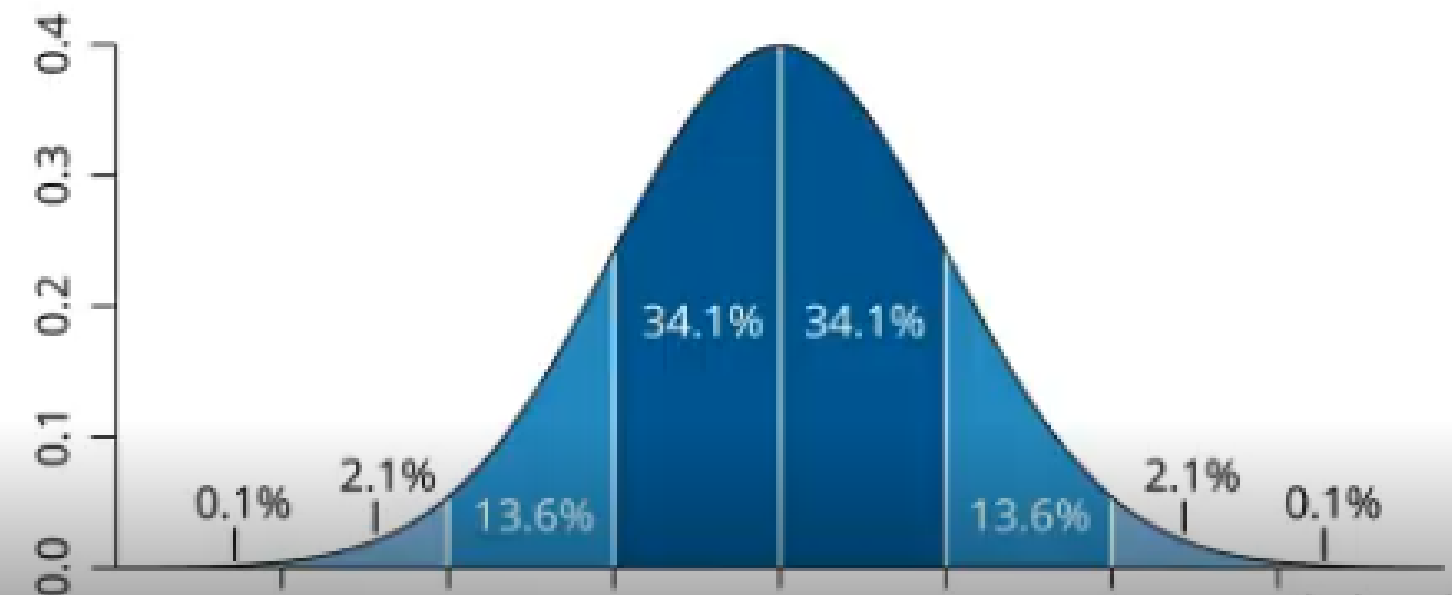
Measure Of Variability



→ Standard Deviation

Standard deviation is the square root of the variance. It measures how spread out the data is around the mean in the same units as the original data.

$$\sigma = \sqrt{\sigma^2}$$



SOLVE THIS QUESTION

Given a dataset of students' scores:

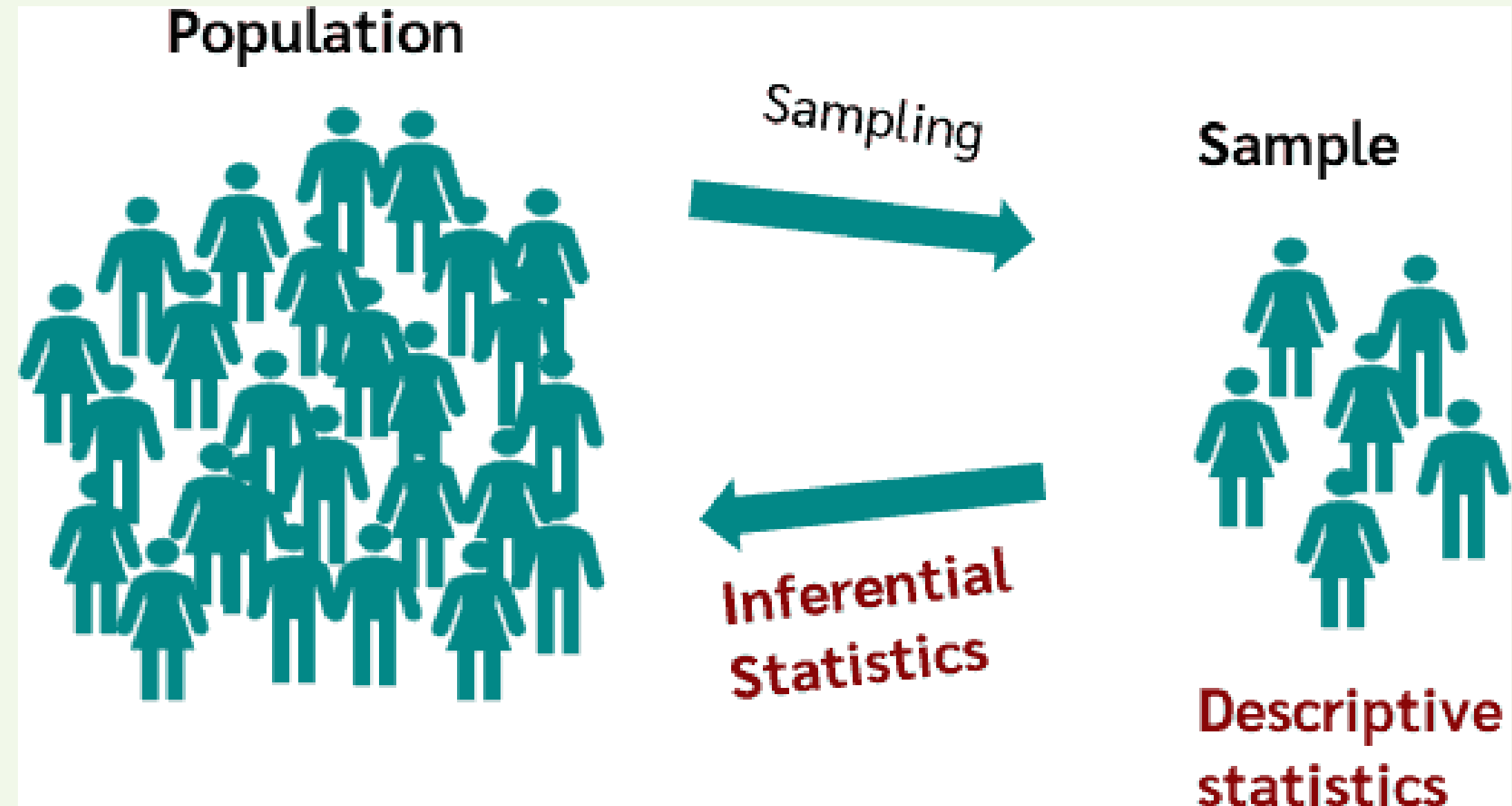
$\{85, 90, 78, 92, 88, 76, 95, 89, 84\}$

1. Calculate the mean, median, and mode.
2. Determine the range, variance, and standard deviation.

Inferential Statistics

- Inferential Statistics is a branch of Statistics used to make predictions, conclusions, or decisions about a large population using a small sample
- Instead of asking all people, we study a small group and infer conclusions.
- relation between 2 features : smoking & Body Fat Percentage

Is there a statistical relationship between smoking and fat?



inferential statistics :

ناخذ عينه صغيرة و نستخدمها علشان نتوقع حاجه عن مجموعة كبيرة
شخص في مصر 100 M

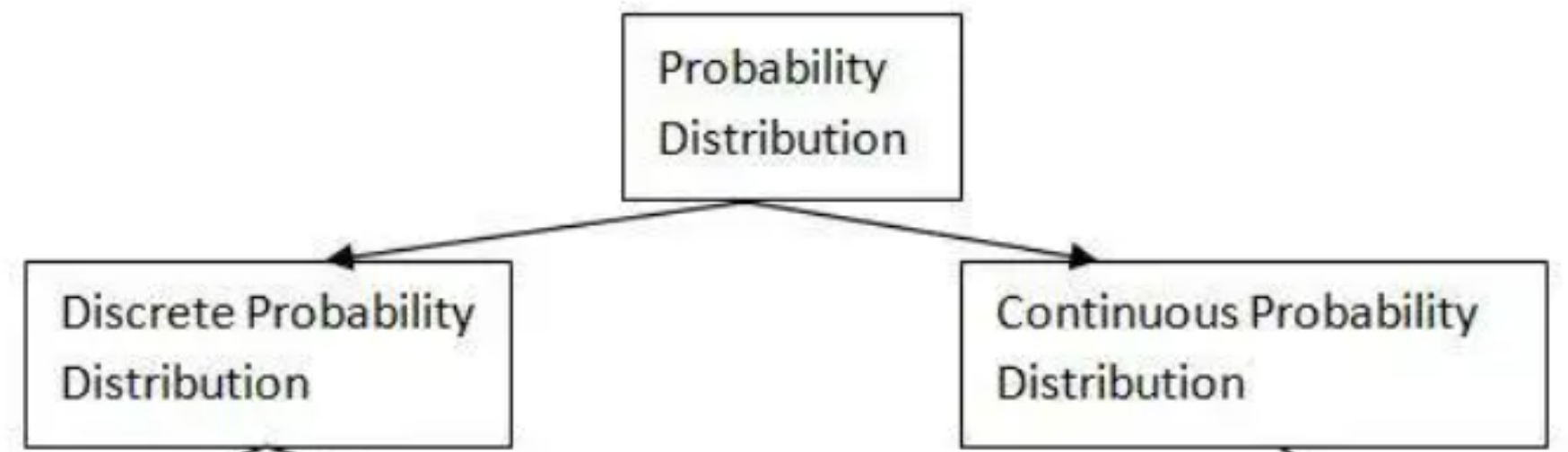
عايزة ادرس نظرية معينه هل التدخين بياثر على نسبة الدهون في الجسم
هختار بس 1000 شخص و اجمع بياناتهم و ادرس و اشوف العلاقة بين التدخين و نسبة الدهون 1-
بعد كدا هستنتج النتيجة سواء التدخين ليه علاقه بالسمنه او لا 2-
و دا اسمه inferential statistics ليه ؟
لاني استنتجت حاجه عن عدد كبير من الناس باستخدام عينه صغيرة

Distributions

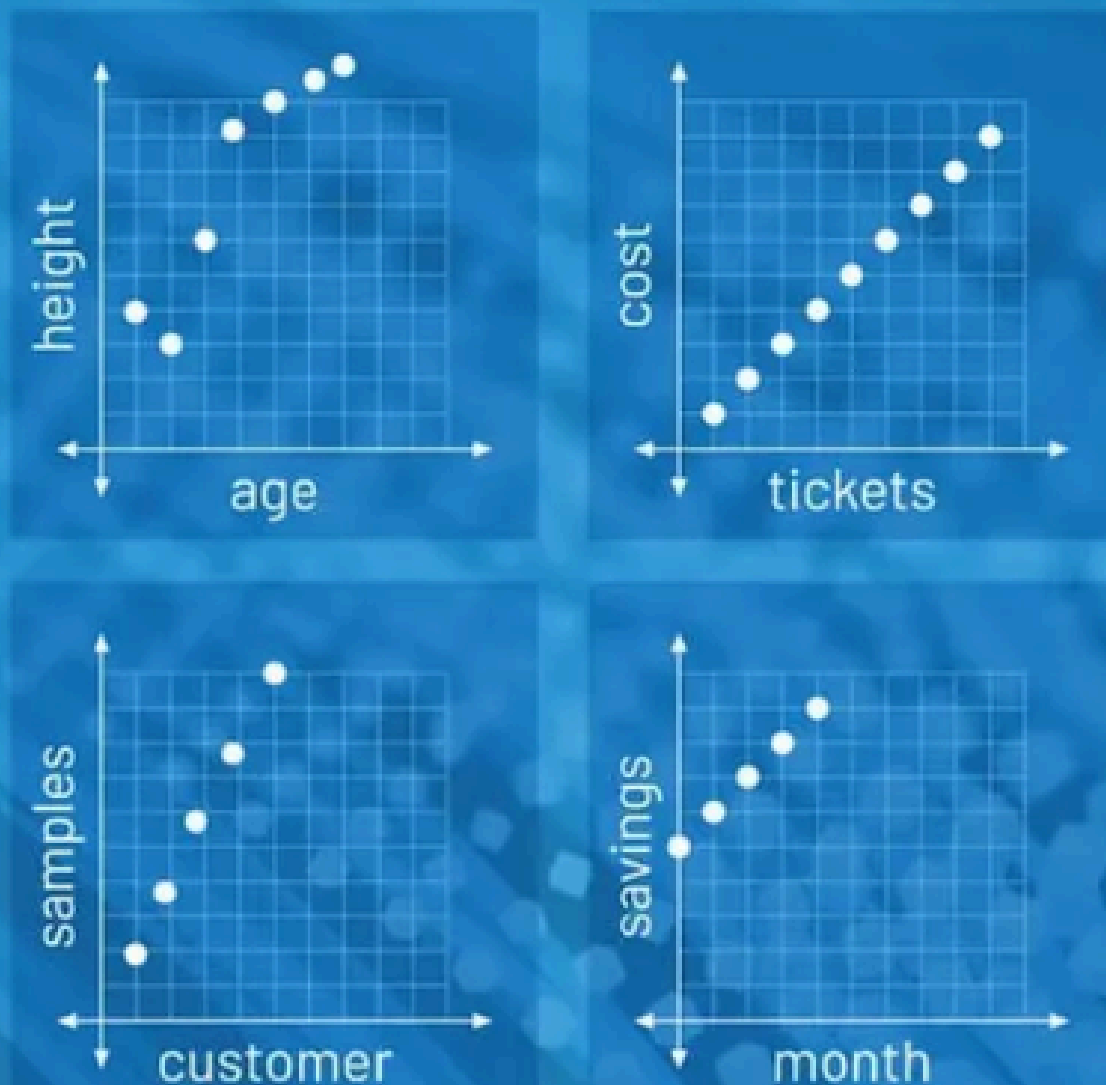
- types of Probability Distributions

Discrete → counting things

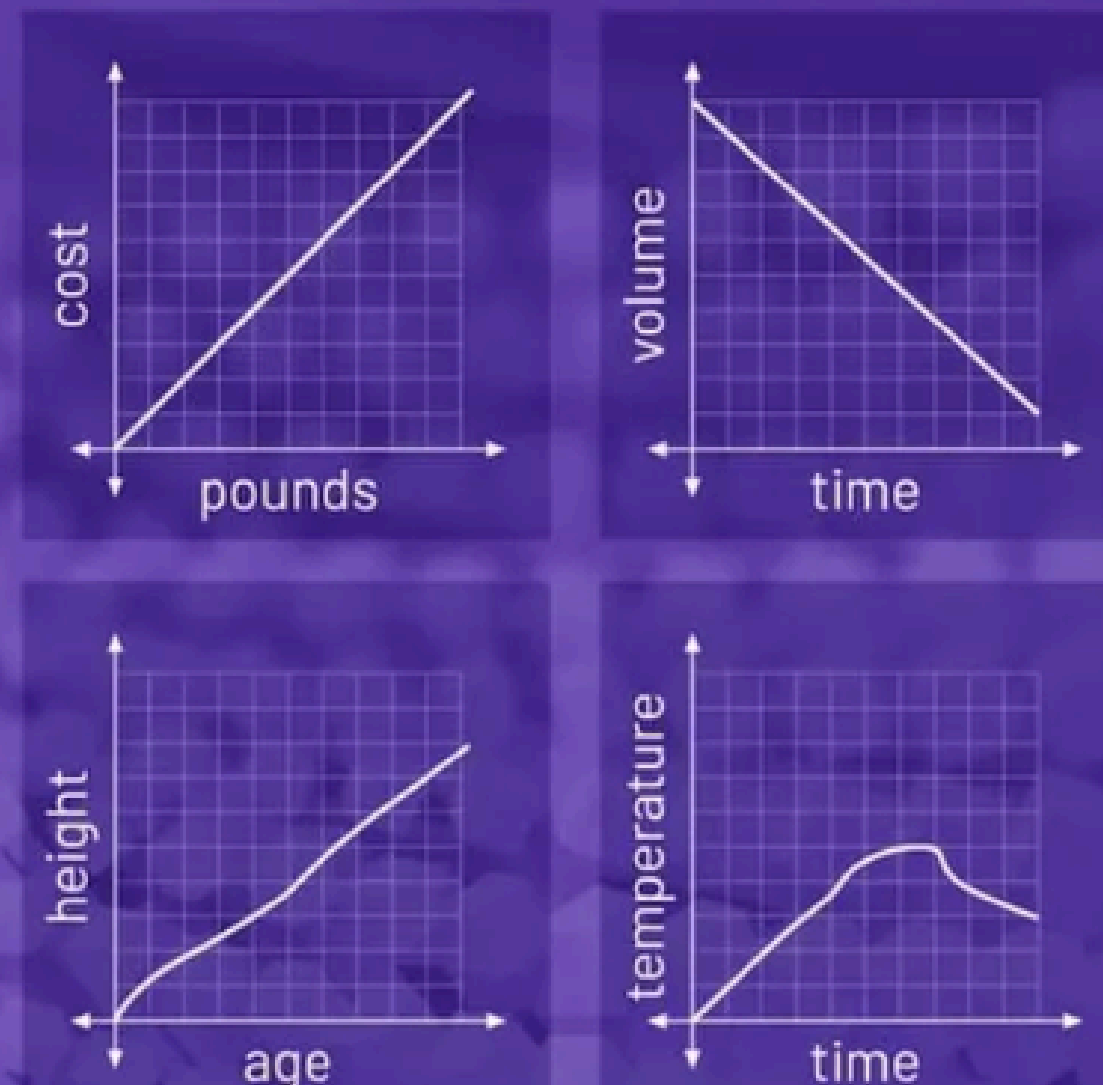
Continuous → measuring things



DISCRETE



CONTINUOUS



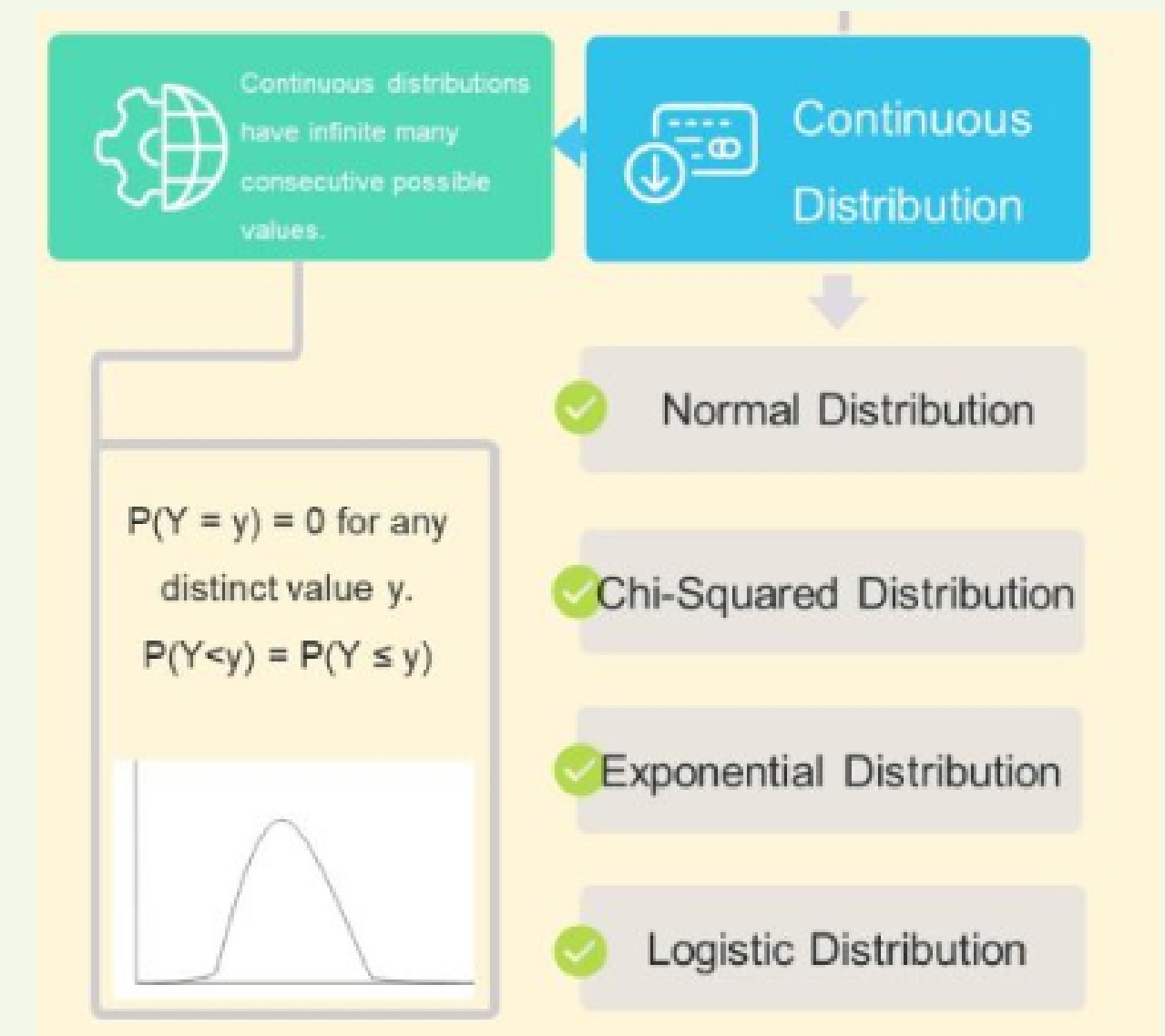
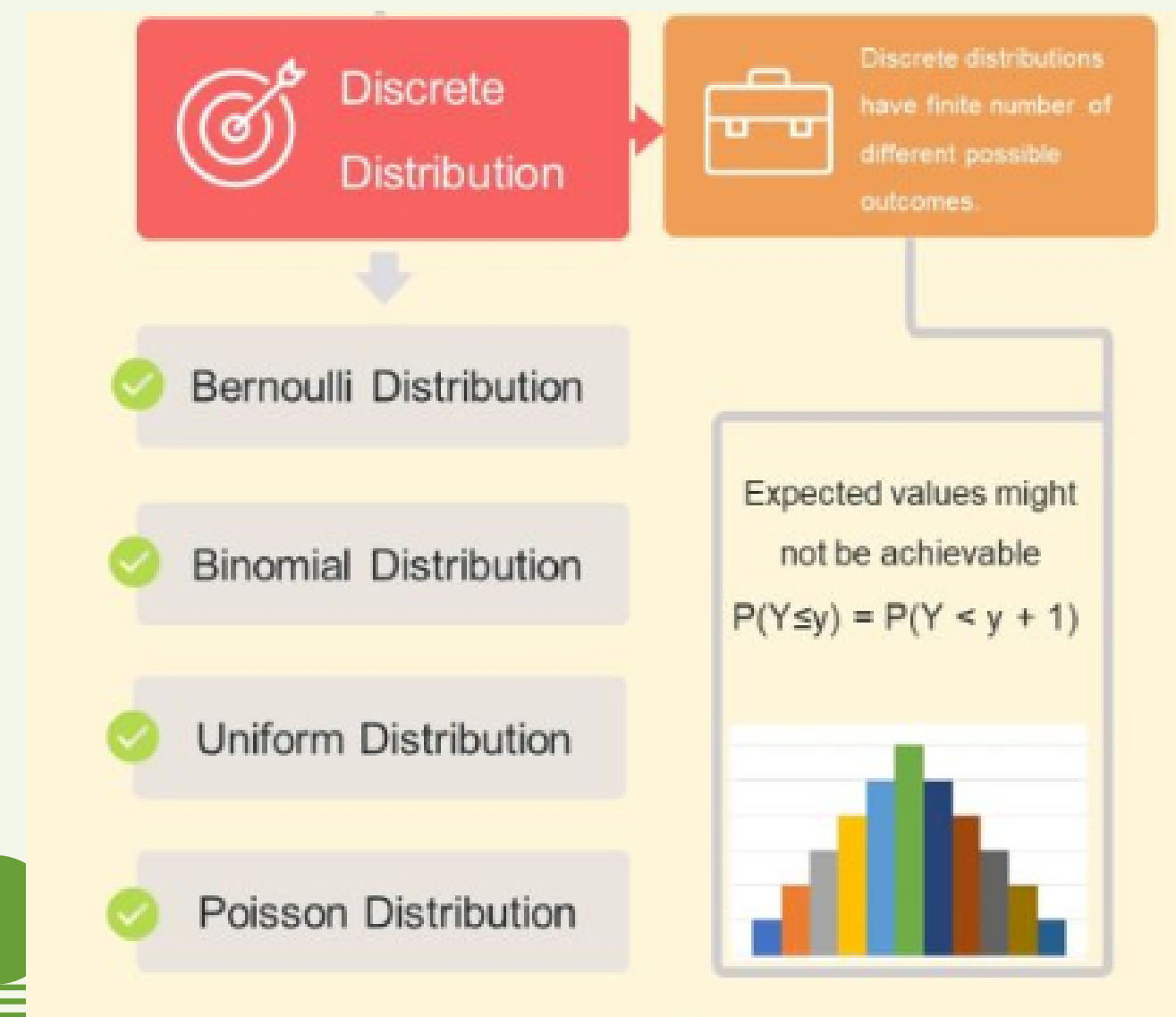
Continuous → infinite numbers can't count it, but extract many insights from these numbers

Distributions

Discrete

These are used when the values are countable (like 0, 1, 2, 3...).

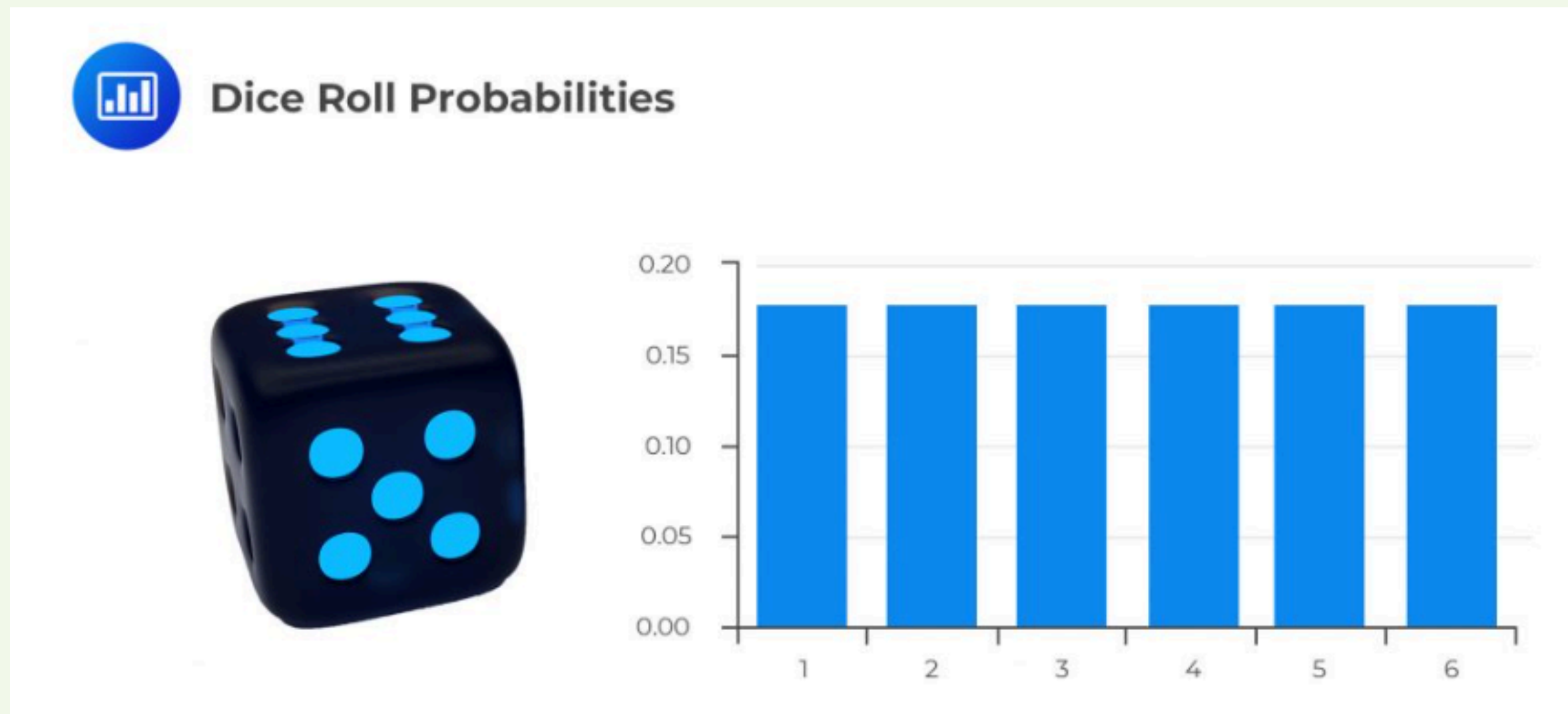
Continuous



Discrete Distributions

Uniform Distribution

All values within the specified range have the same chance of occurring. All outcomes are Equally likely



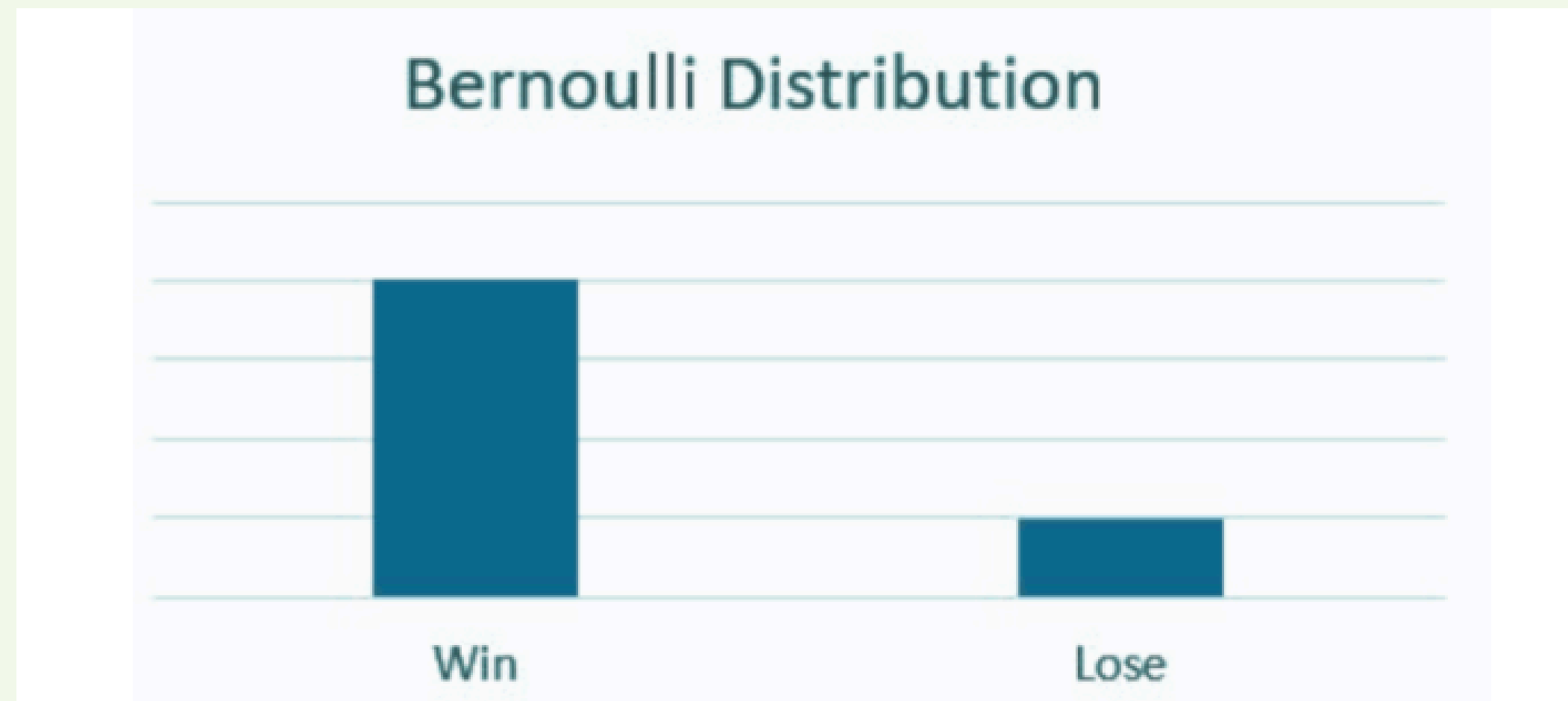
Discrete Distributions

Bernoulli Distribution

It is a distribution that describes an experiment with only two outcomes:

Success = 1

Failure = 0



Discrete Distributions

Binomial Distribution

Binomial = Many Bernoulli experiments

Poisson Distribution

“How often something happens” **within a specific time period** or space.
event may happened , may not happened

two events can't take place at the same time (البرق و الرعد)

Continuous Distributions

Normal Distribution

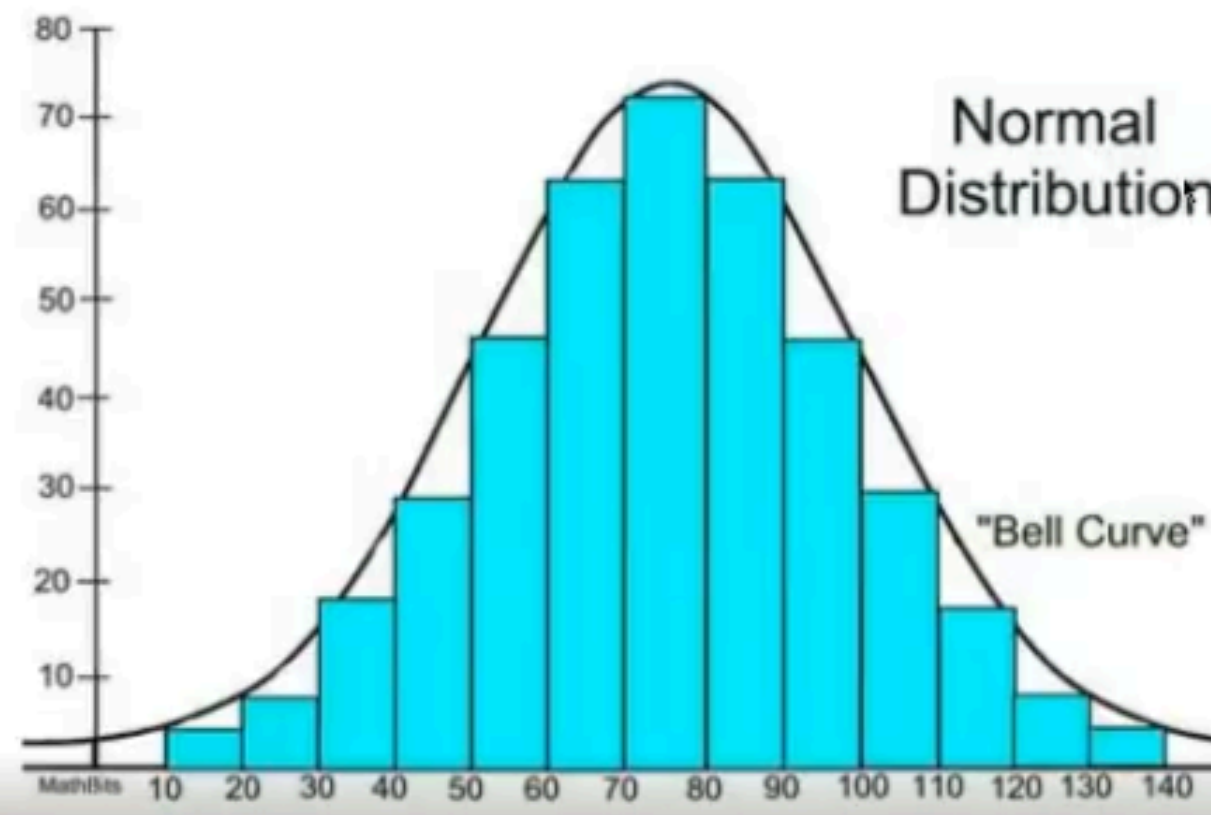
range numbers in standard Normal Distribution $-3.5, 3.5$

probability under this curve is 99.998

STD = 1

mean = 0

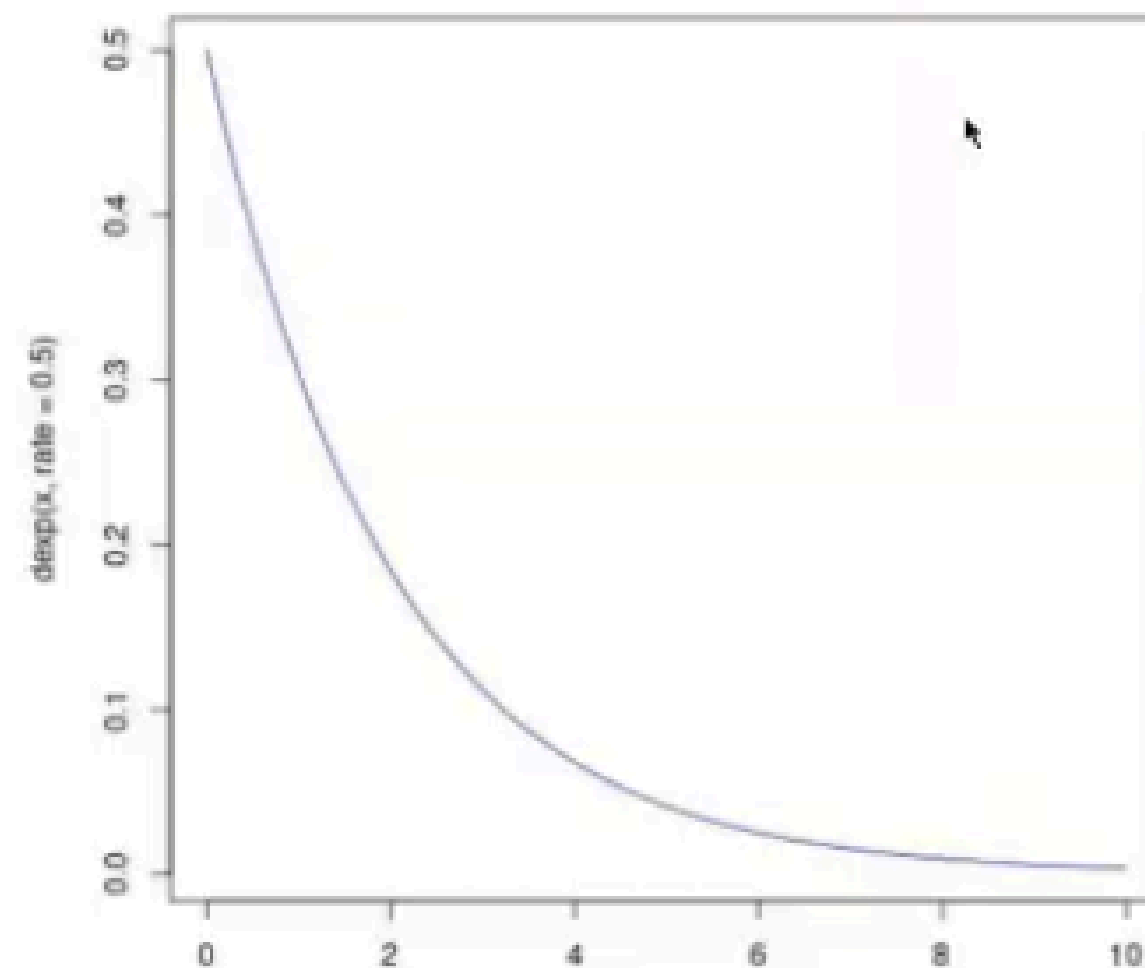
Normal Distribution: Symmetric Distribution of Values Around the Mean



Continuous Distributions

Exponential Distribution

exponential Distribution: Model Elapsed Time between Two Events

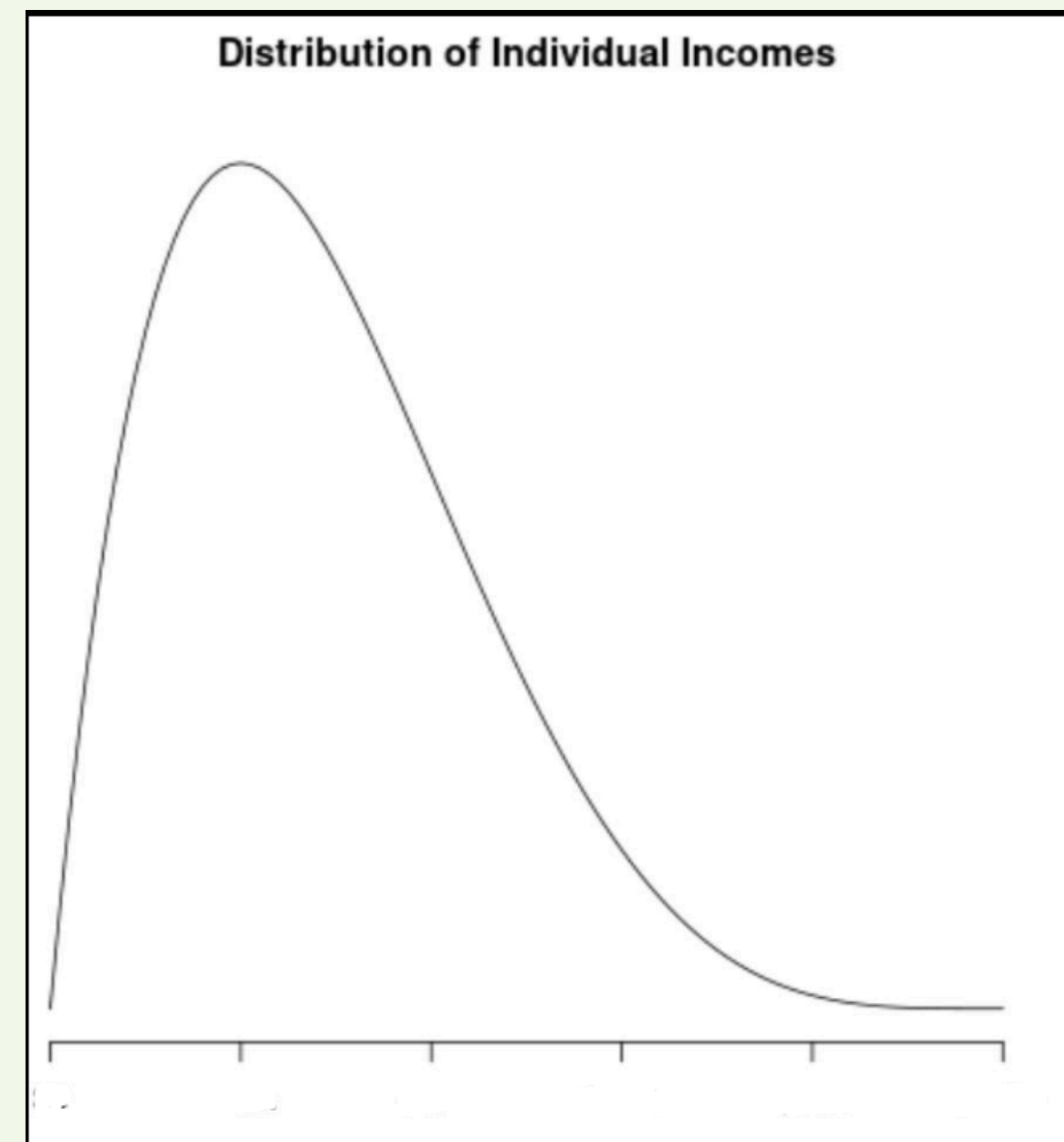
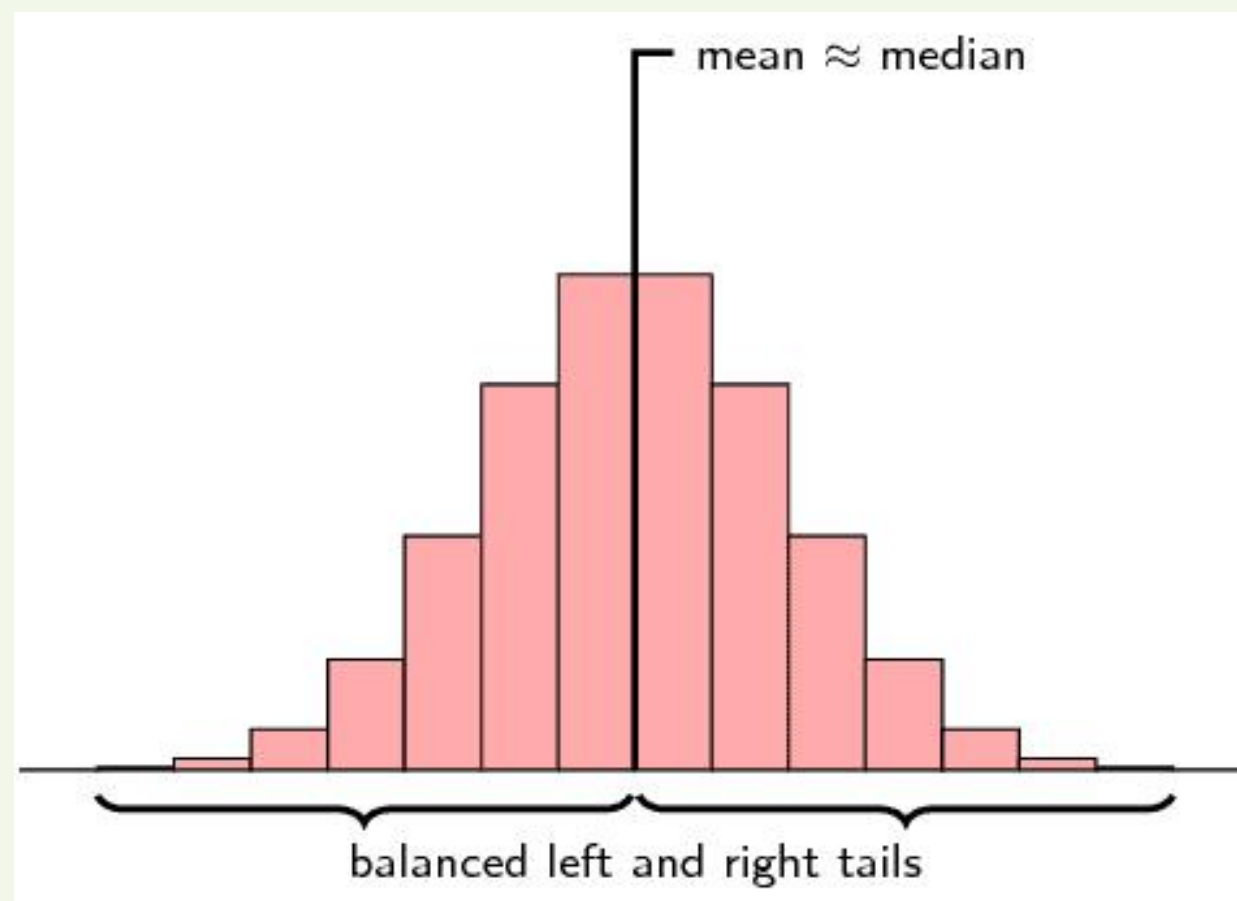


نسبة التطور بين 2
Event بالنسبة للوقت
كل لما بنكبر صحتنا
بتقل

[3000, 3500, 4000, 4500, 5000, 70000]

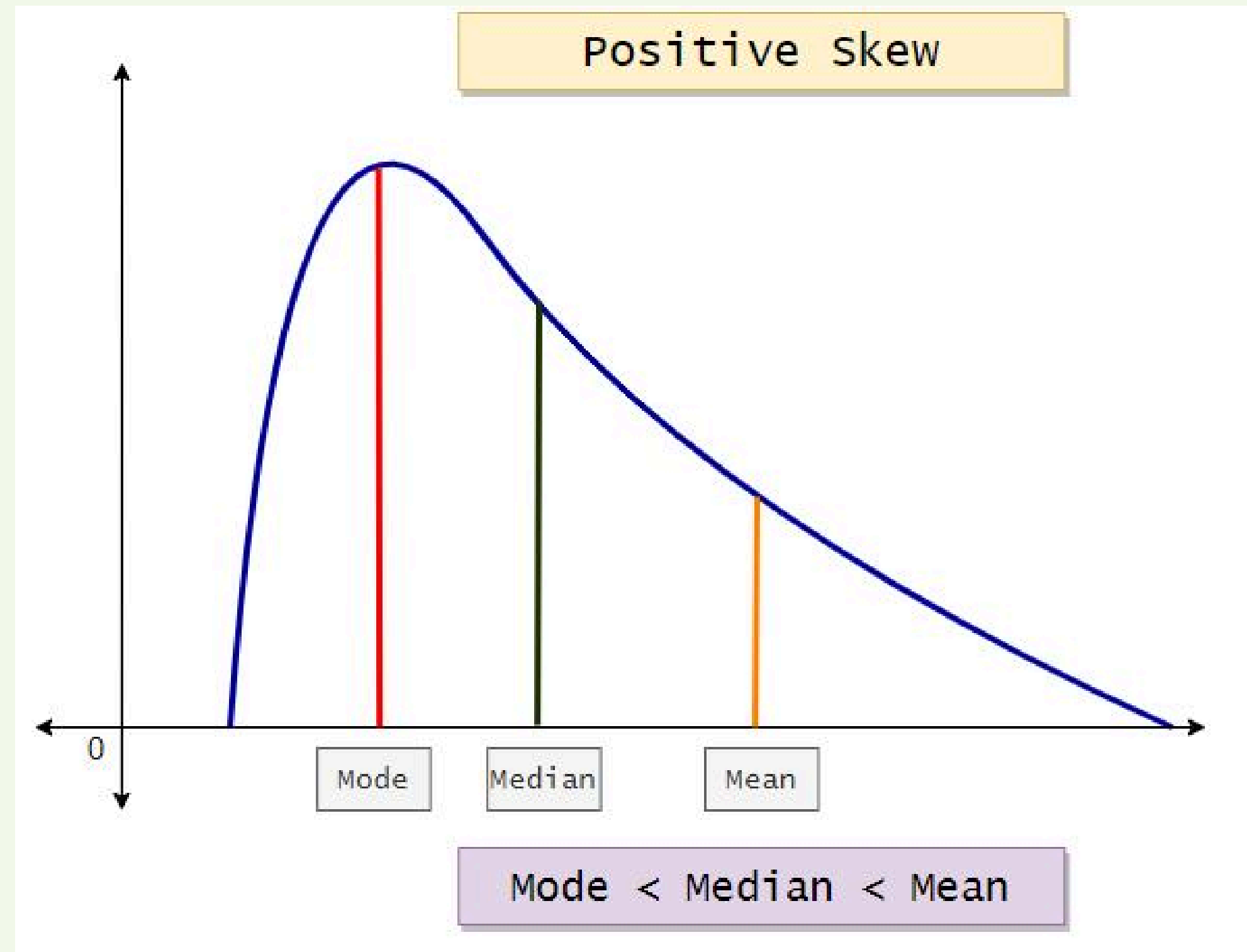
Normal Distribution (No Skew)

Salary [5k , 3k] , [70k]



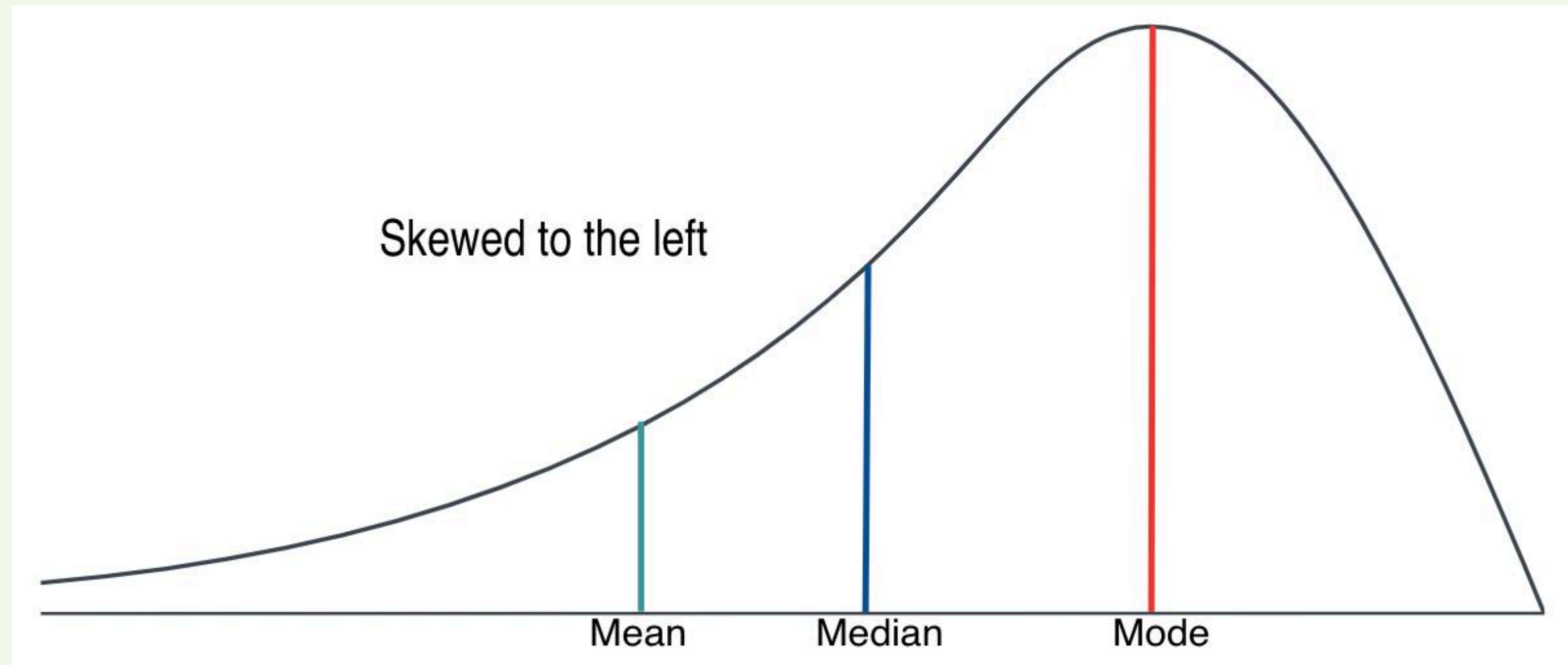
[3000, 3500, 4000, 4500, 5000, 70000]

Mean = 15,000 **X**
Median=4250
Mean>Median

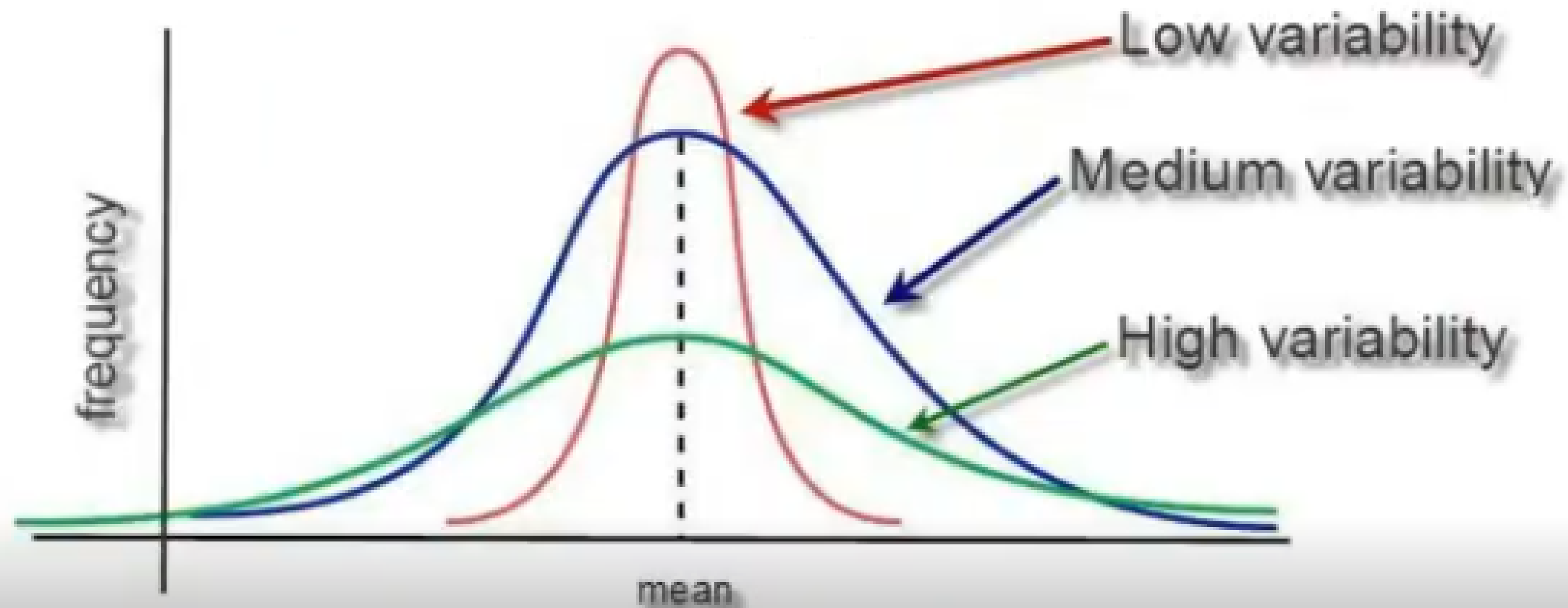


[20, 85, 90, 92, 95, 98]

Mean = 80
Median = 91
Mean < Median

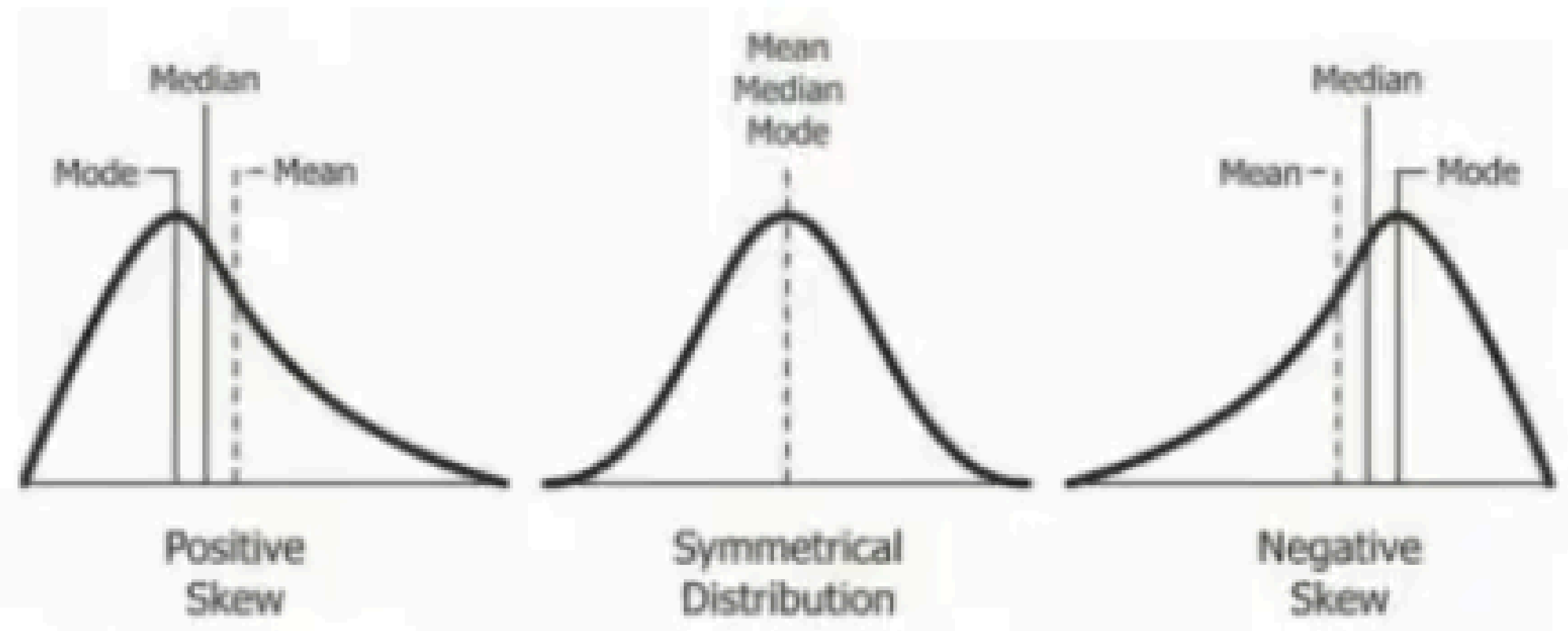


3 curves Normal Distribution
mean = 0
but std $\neq$ 1



SKEWED DATA

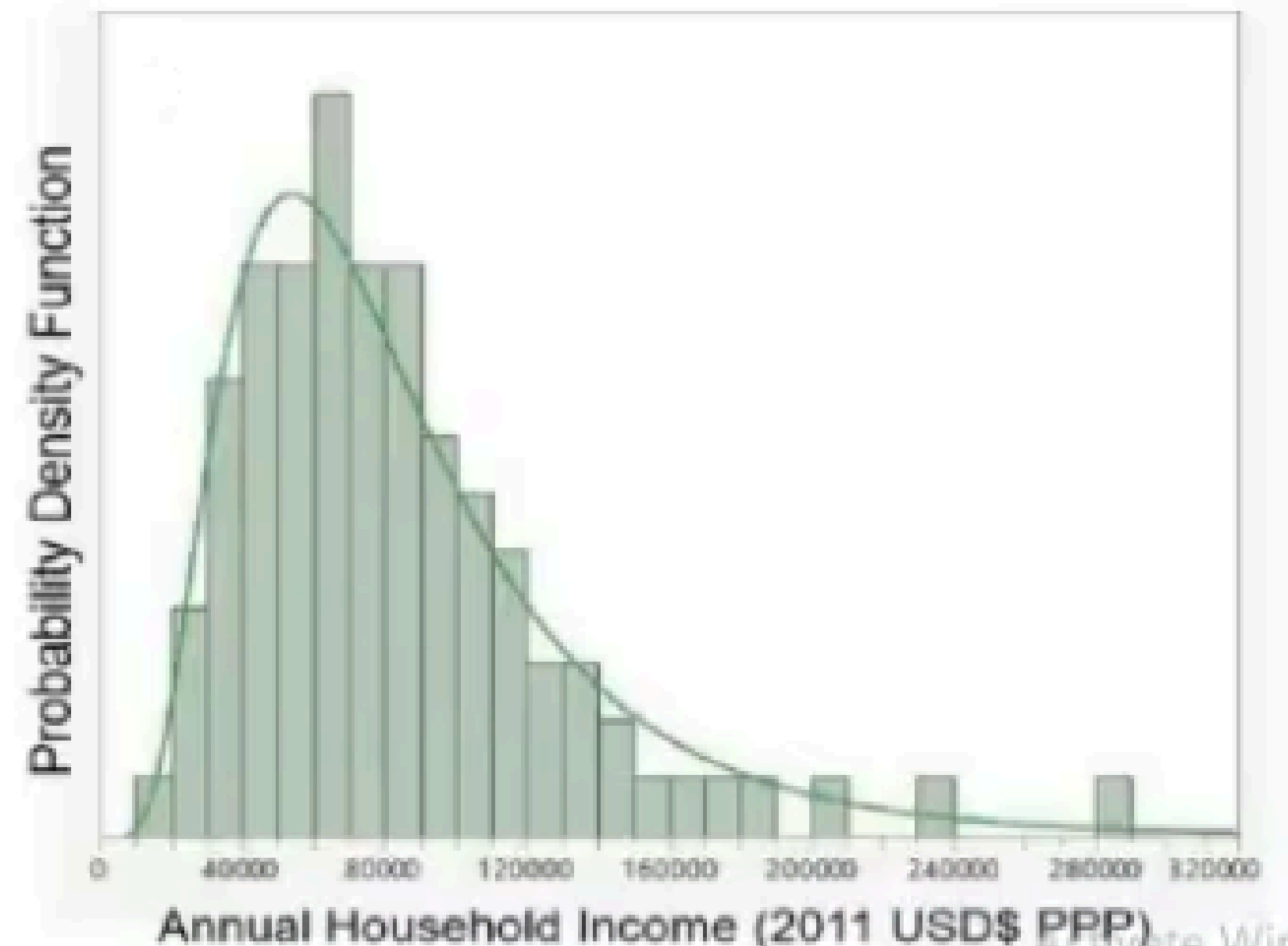
Skewed data refers to a dataset where the distribution of values is asymmetrical, with the tail of the distribution extending towards one side. The curve appears distorted or stretched in one direction, either to the left or right.



SKEWED DATA

Positively Skewed (Right-Skewed)

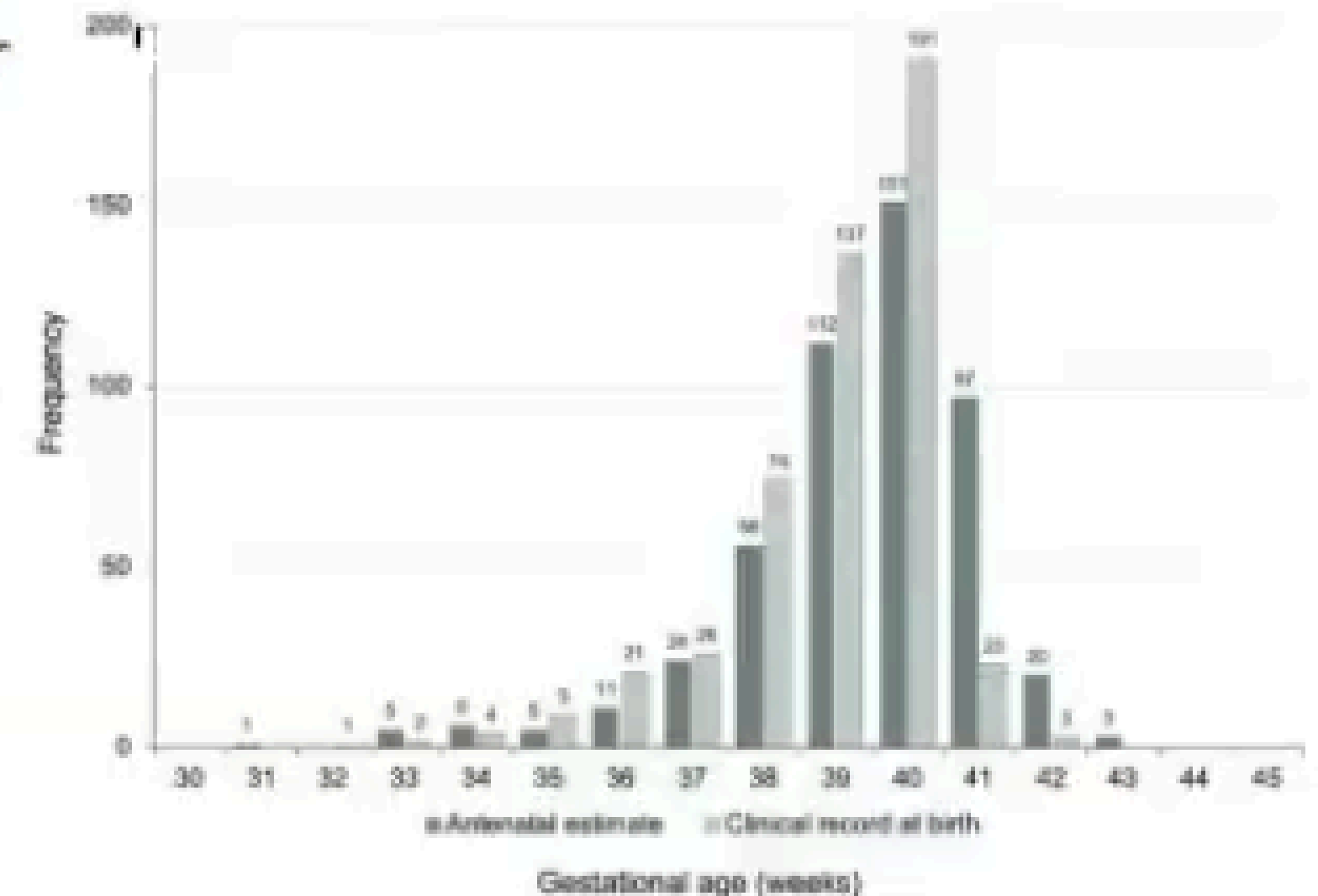
- The tail of the distribution is longer on the right side.
- Most of the values are concentrated on the lower end, and a few higher values (outliers) stretch the distribution.
- Examples: Income distributions, house prices.
- $\text{Mean} > \text{Median} > \text{Mode}$.



SKEWED DATA

Negative Skewness (Left Skewed)

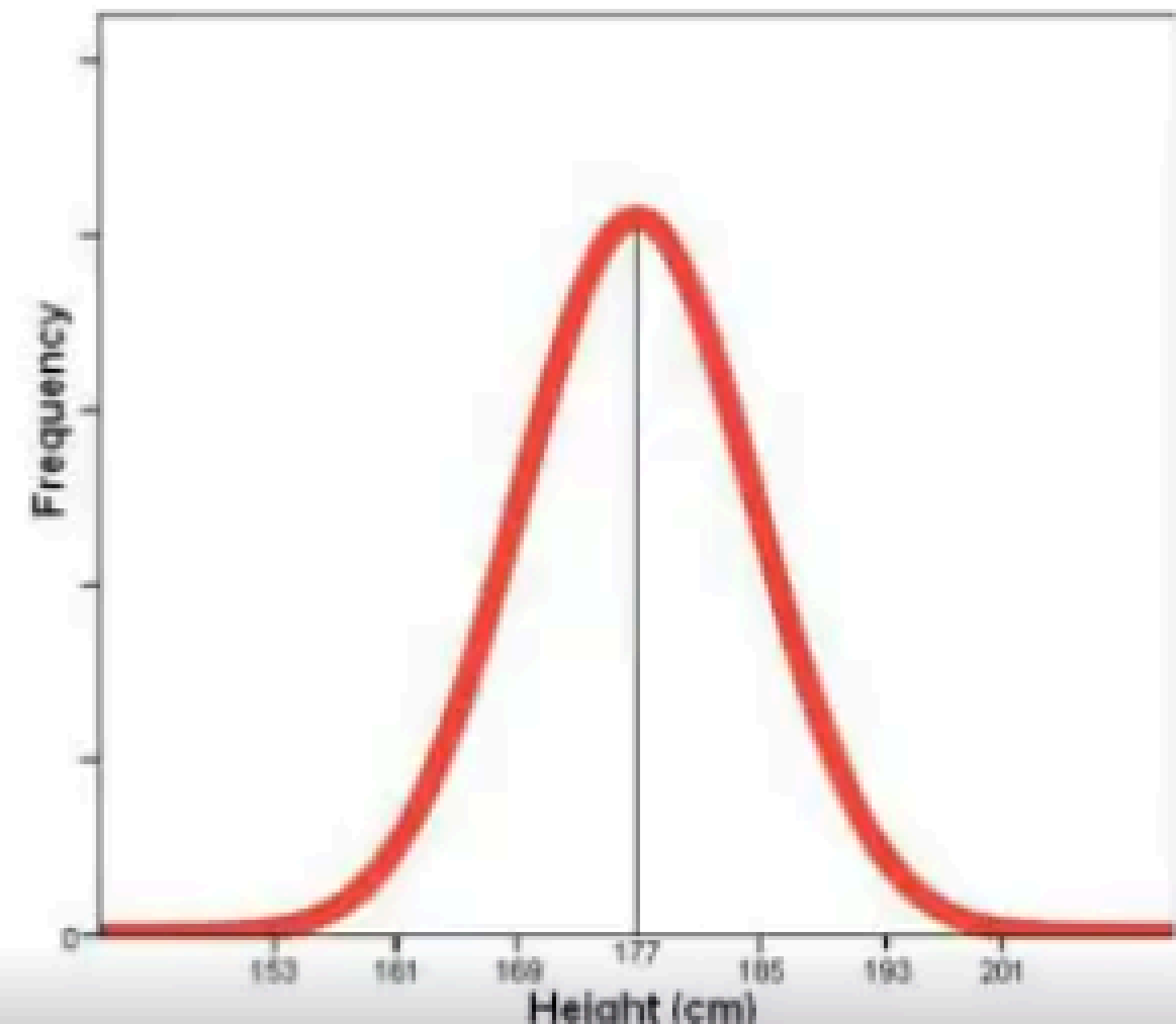
- The tail of the distribution is longer on the left side.
- Most of the values are concentrated on the higher end, and a few lower values stretch the distribution.
- Examples: Age at retirement, exam scores in easy tests.
- $\text{Mean} < \text{Median} < \text{Mode}$.



SKEWED DATA

Zero-Skewed Data Distribution

- The data is evenly distributed around the mean.
- Example: A perfectly normal distribution.



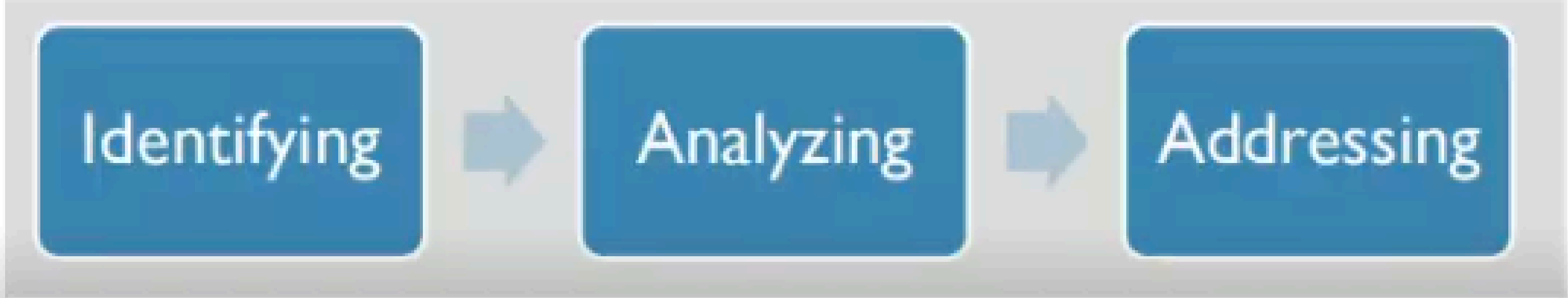
SKEWED DATA

Implications of Skewed Data

- **Positively Skewed Data:**
 - Mean may not represent the central tendency accurately.
 - Analysis may overestimate the "typical" value.
- **Negatively Skewed Data:**
 - Similar issues where the mean may underestimate the central value.

OUTLIER DETECTION

Outlier detection is the process of identifying and removing outliers to improve data accuracy and analysis.



```
graph LR; A[Identifying] --> B[Analyzing]; B --> C[Addressing]
```

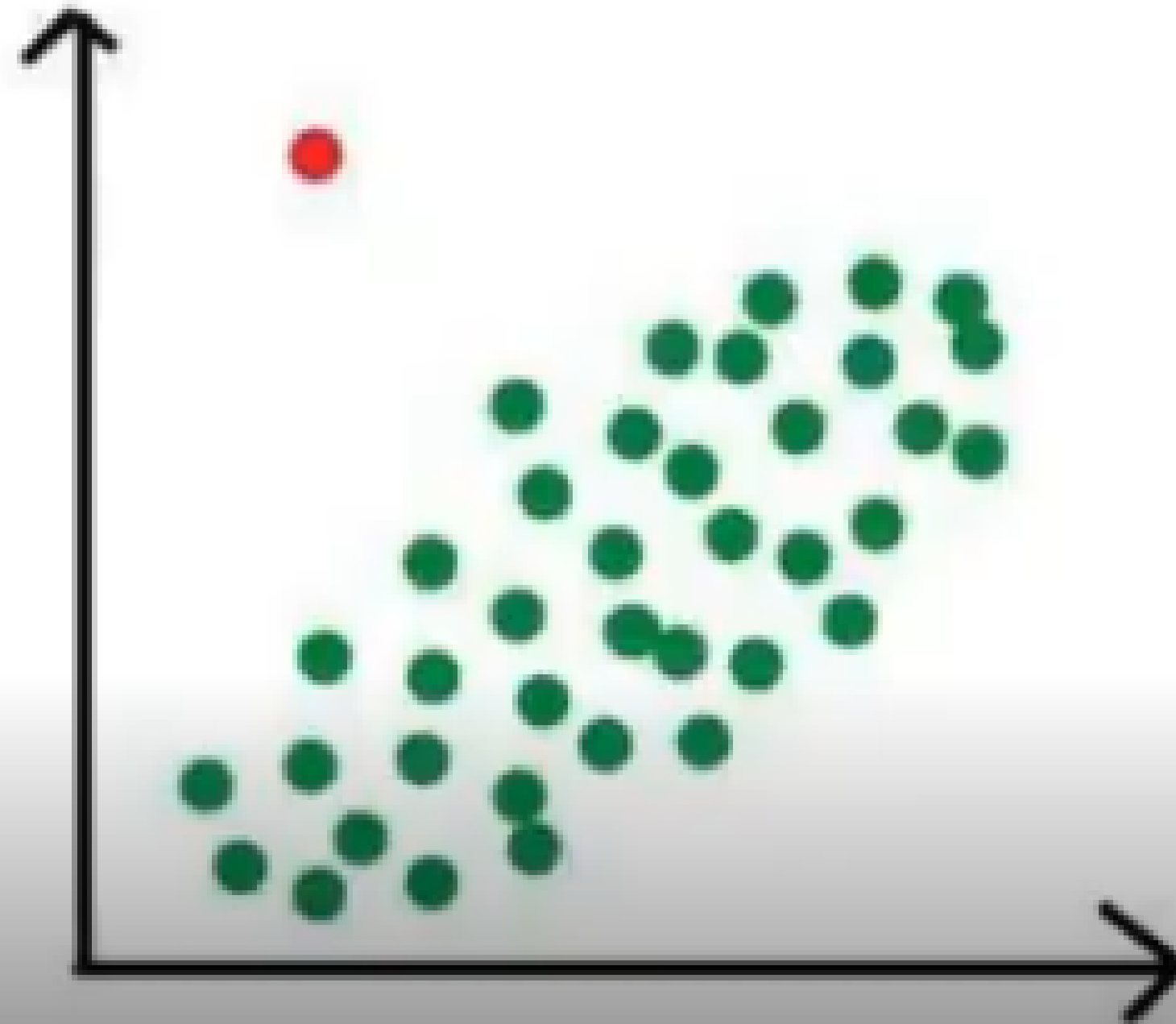
Identifying

Analyzing

Addressing

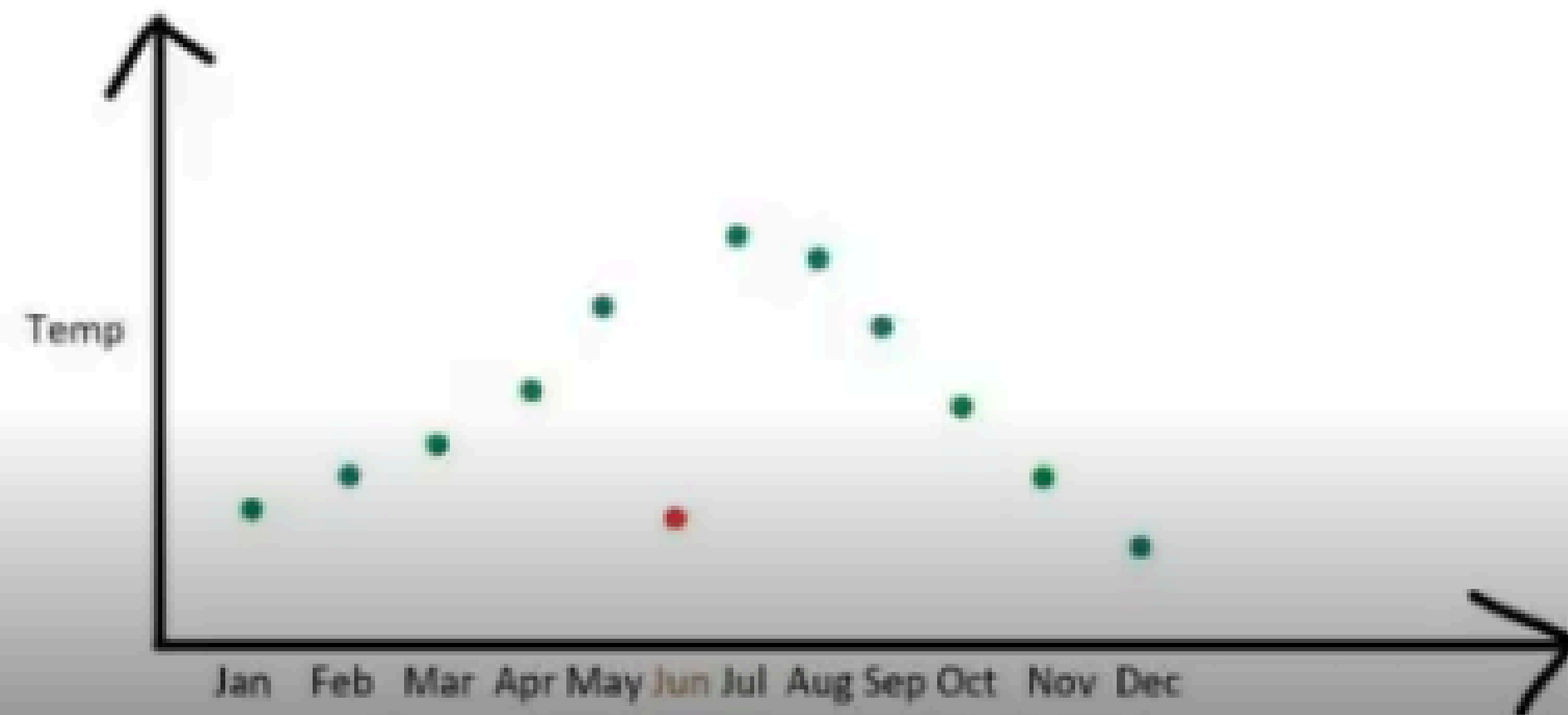
TYPES: GLOBAL OUTLIERS

A global outlier is a data point that is significantly different from all the other data points in the entire dataset.



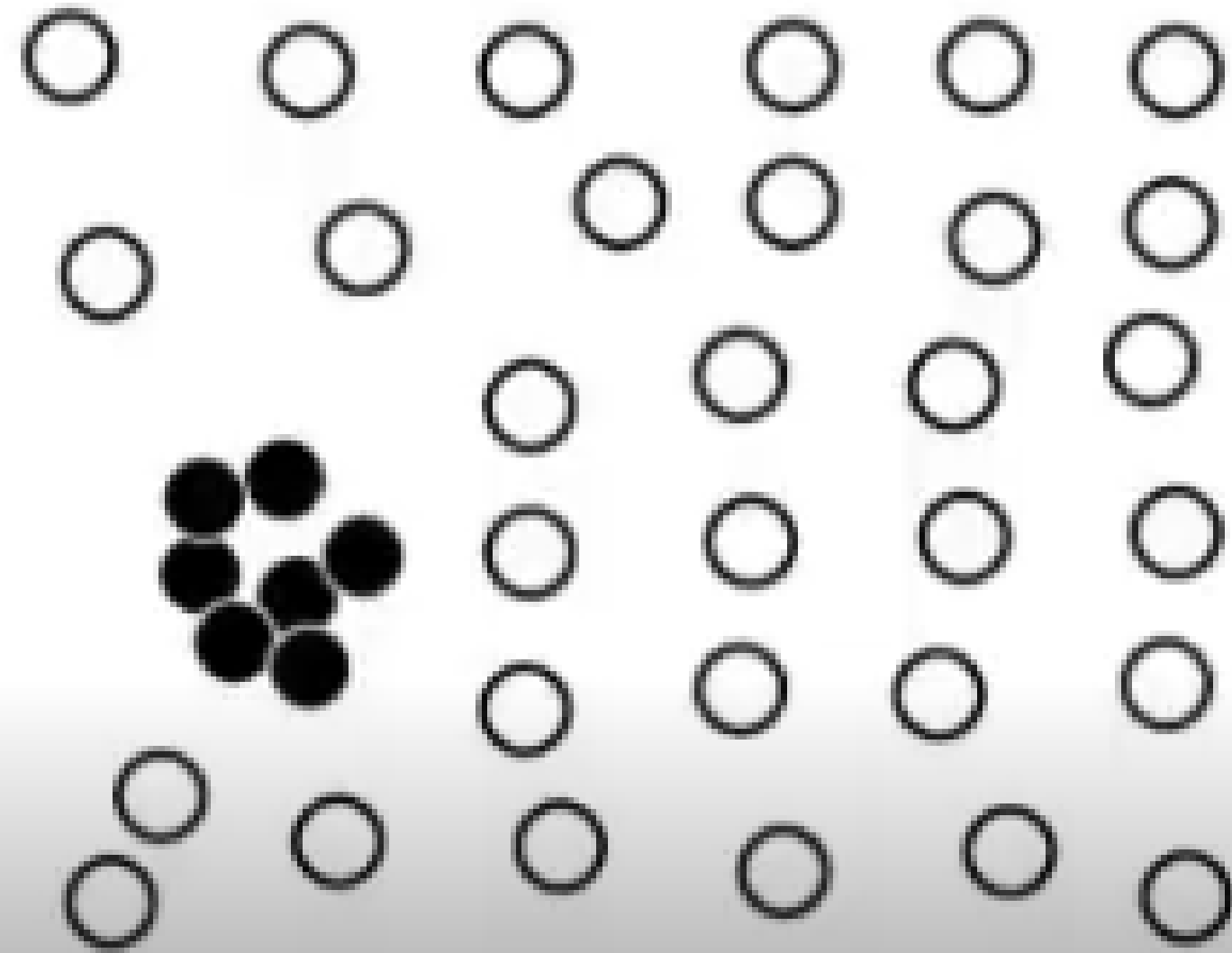
TYPES: CONDITIONAL (CONTEXTUAL) OUTLIERS

Contextual outliers are data points that deviate significantly from the expected behavior within a specific context or subgroup



TYPES: COLLECTIVE OUTLIERS

Collective outliers are groups of data points that collectively deviate significantly from the overall distribution of a dataset.

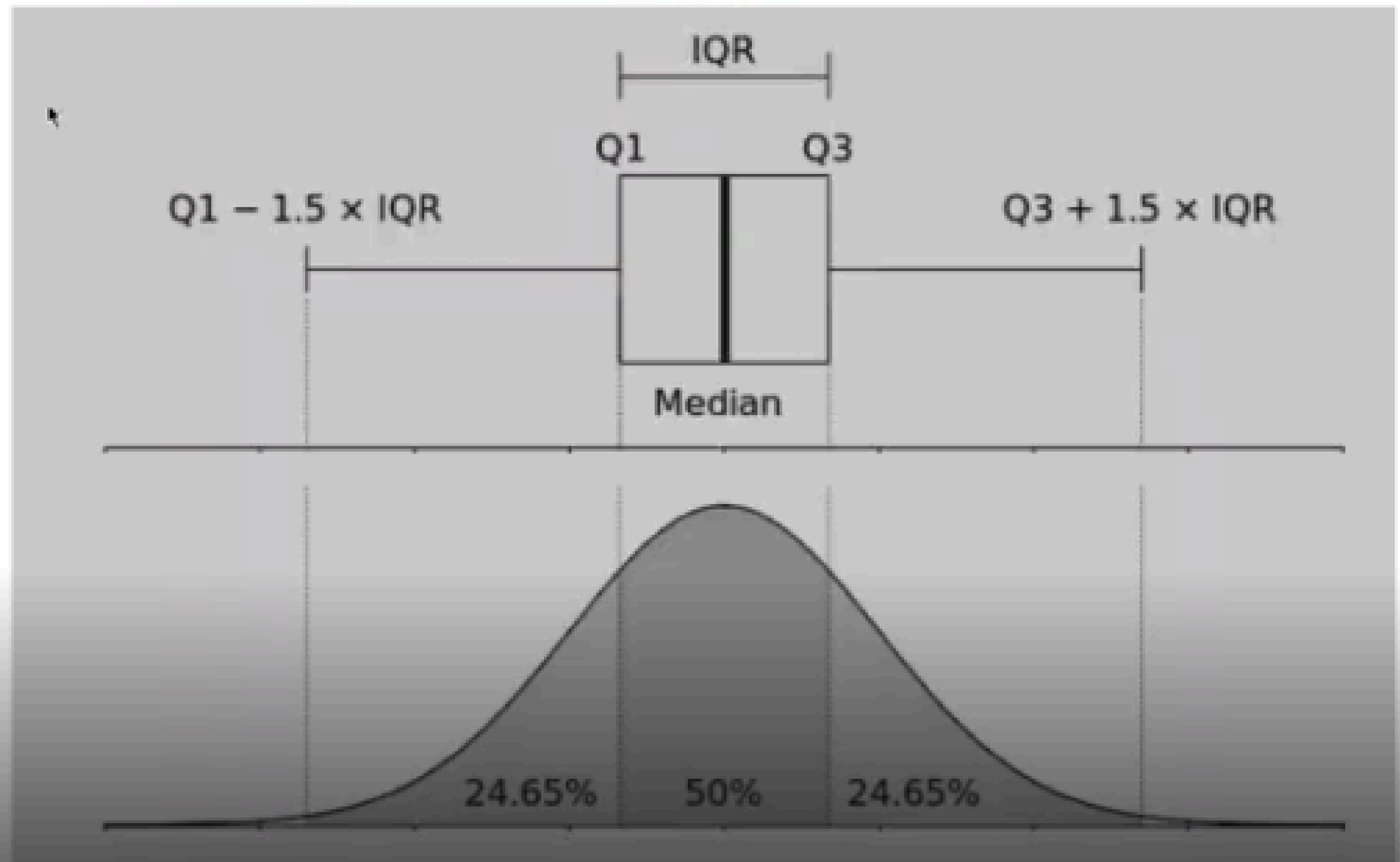


INTERQUARTILE RANGE (IQR)

$$\begin{aligned} IQR &= Q3 - Q1 \\ &= 8.5 - 3.5 \\ &= 5 \end{aligned}$$



INTERQUARTILE RANGE (IQR)



INTERQUARTILE RANGE (IQR)

Data Set:

3, 7, 8, 5, 12, 14, 21, 13, 18, 10

Step 1: Order the Data:

3, 5, 7, 8, 10, 12, 13, 14, 18, 21

Step 2: Find the Quartiles:

.Q1 (First Quartile): Median of the first half of the data (3, 5, 7, 8, 10) = 7

.Q3 (Third Quartile): Median of the second half of the data (12, 13, 14, 18, 21) = 14

Step 3: Calculate the IQR:

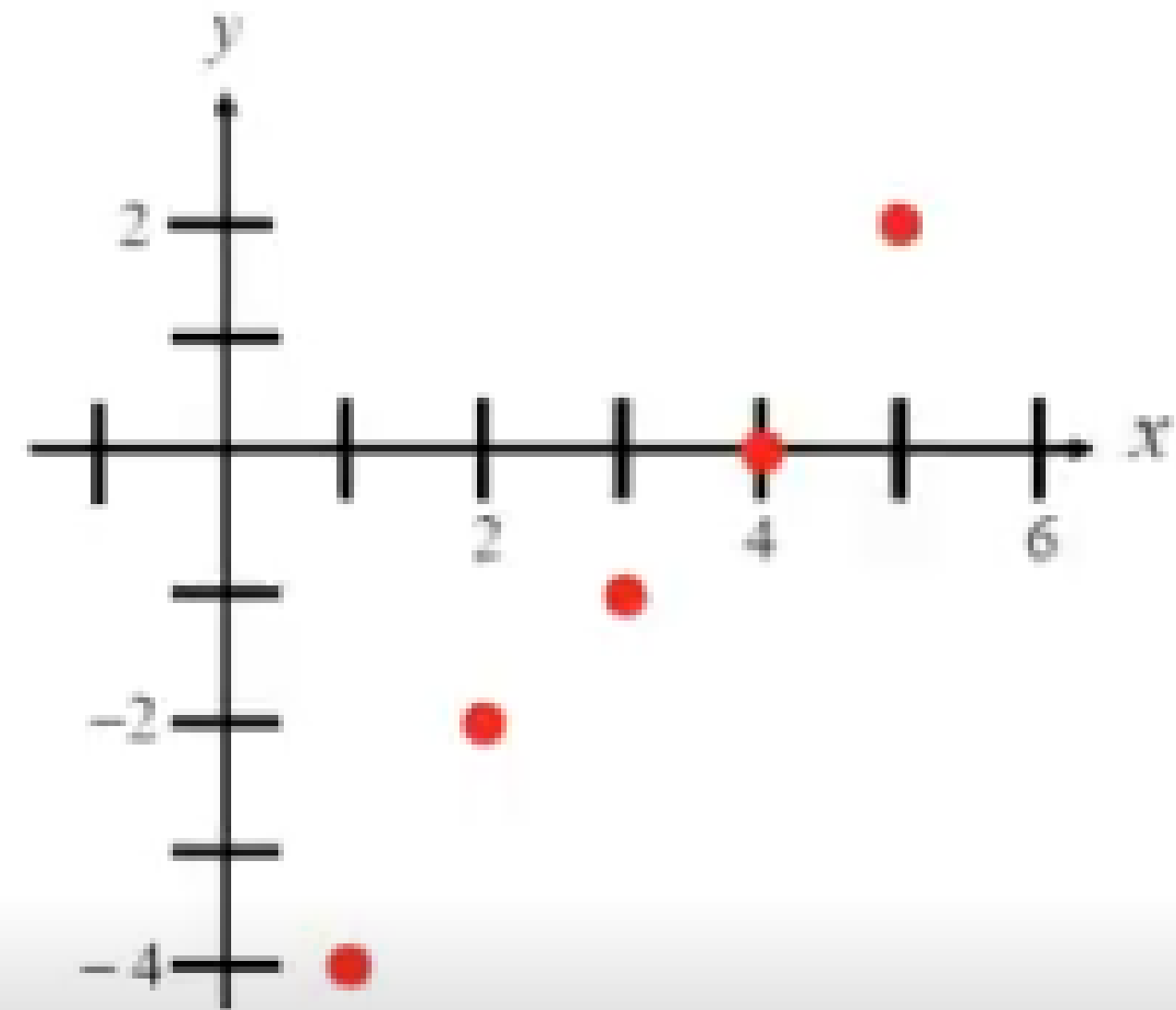
Step 4: Define Outlier Boundaries:

CORRELATION

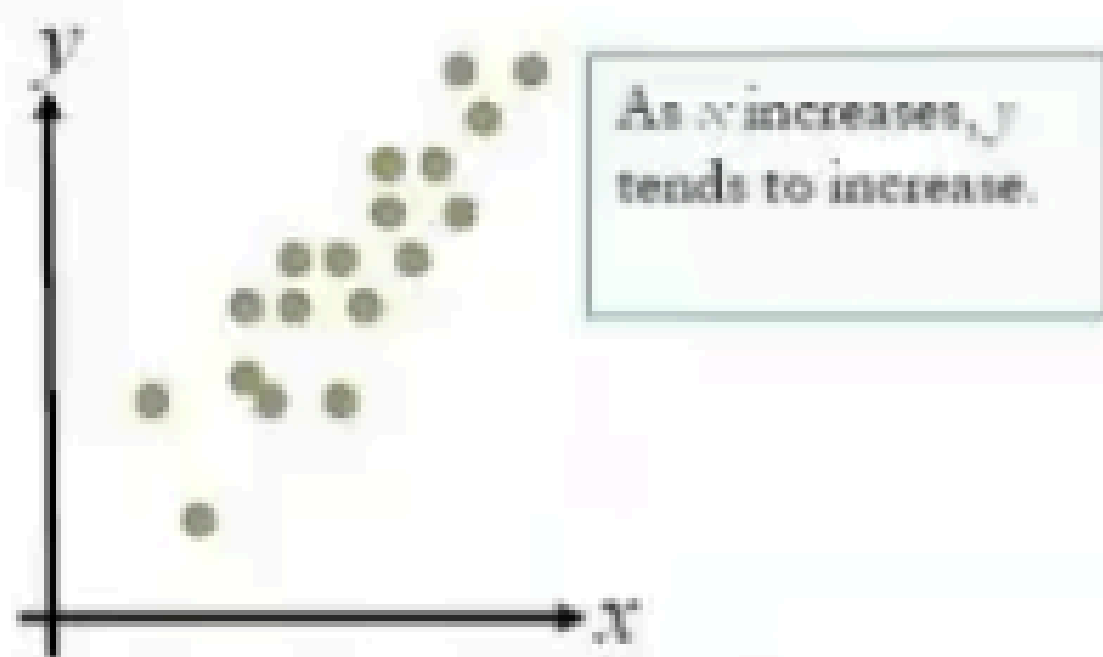
- A **correlation** is a relationship between two variables. The data can be represented by the ordered pairs (x, y) where x is the **independent variable**, and y is the **dependent variable**.
- The independent variable is the **cause**. Its value is *independent* of other variables in your study.
- The dependent variable is the **effect**. Its value *depends* on changes in the independent variable.
- A **scatter plot** can be used to determine whether a linear (straight line) correlation exists between two variables.

SCATTER PLOT

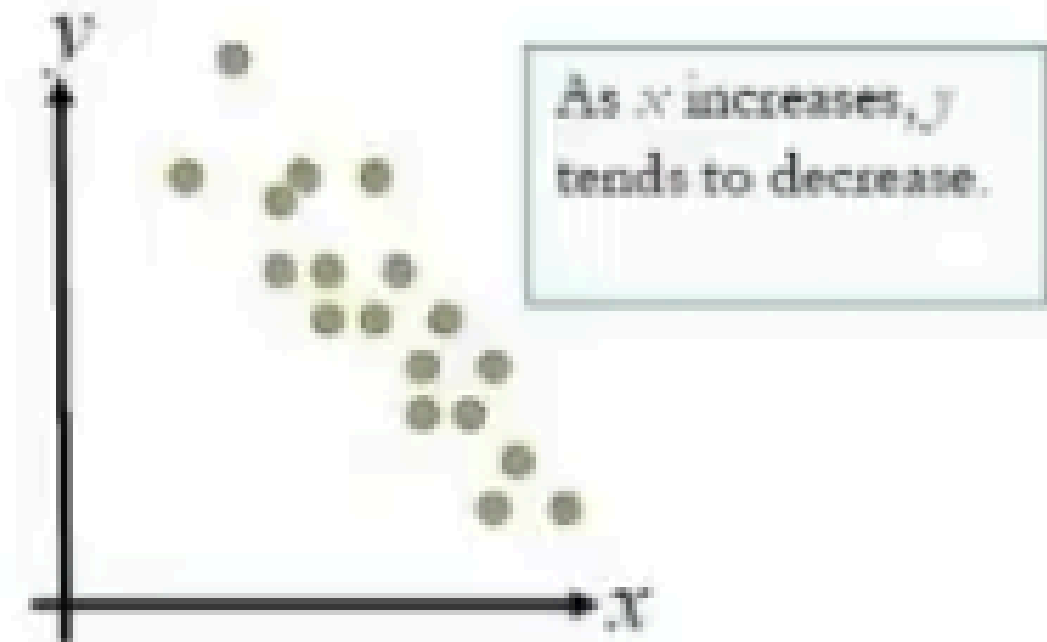
x	1	2	3	4	5
y	-4	-2	-1	0	2



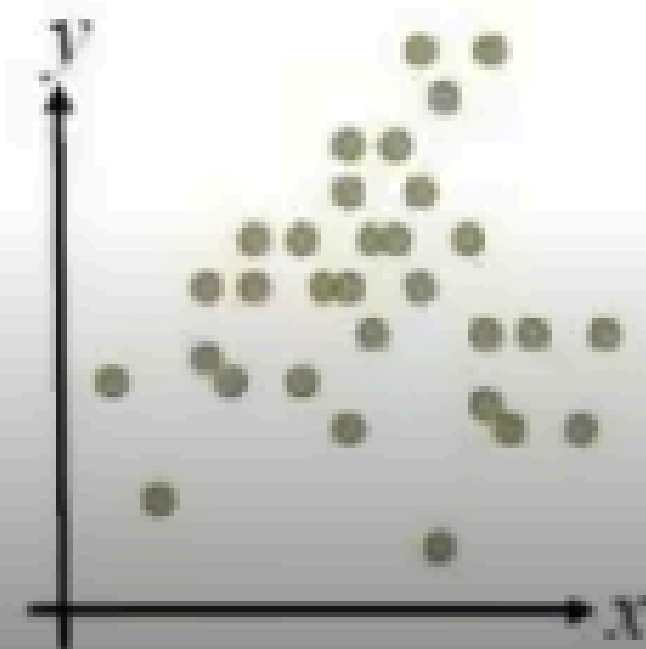
CORRELATION



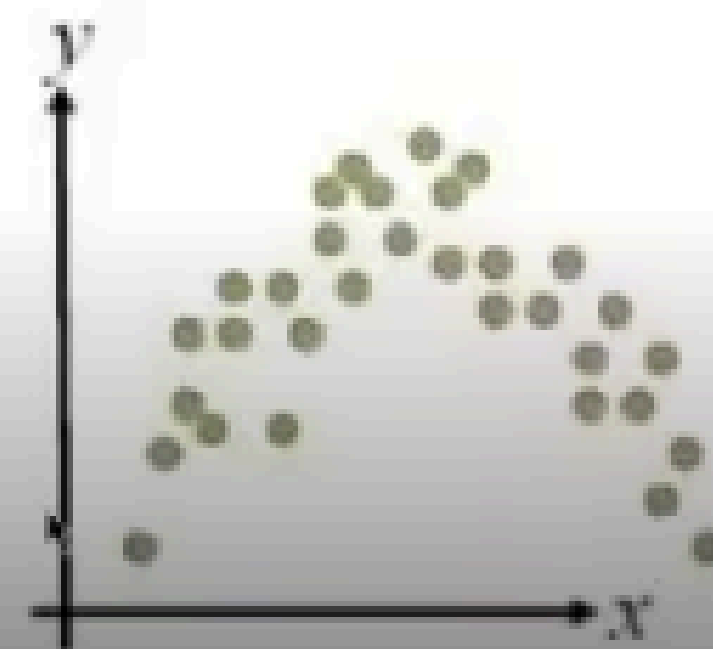
Positive Linear Correlation



Negative Linear Correlation



No Correlation



Nonlinear Correlation

CORRELATION COEFFICIENT

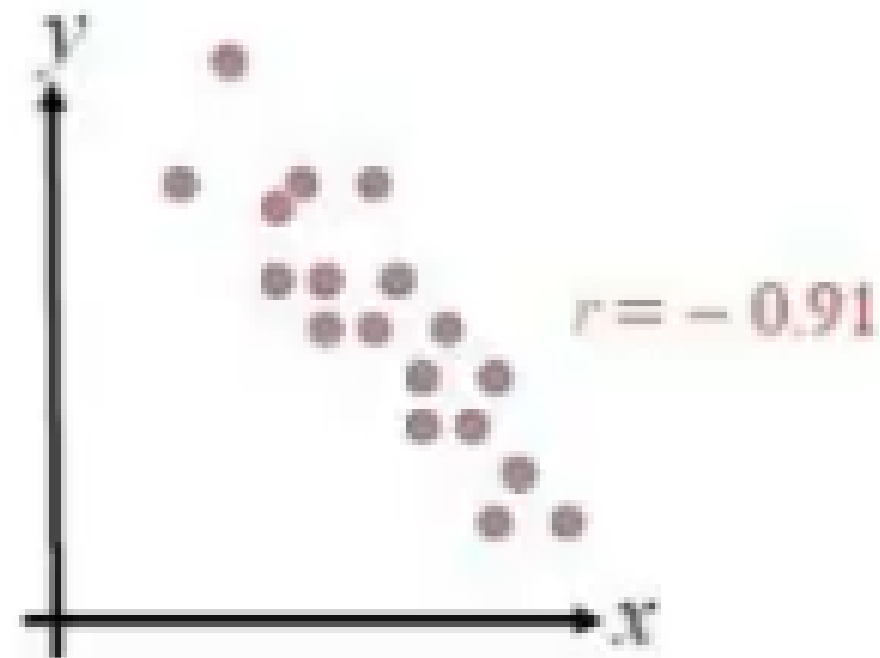
- The **correlation coefficient** is a measure of the strength and the direction of a linear relationship between two variables. The symbol r represents the sample correlation coefficient. The formula for r is

$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \sqrt{n\sum y^2 - (\sum y)^2}}$$

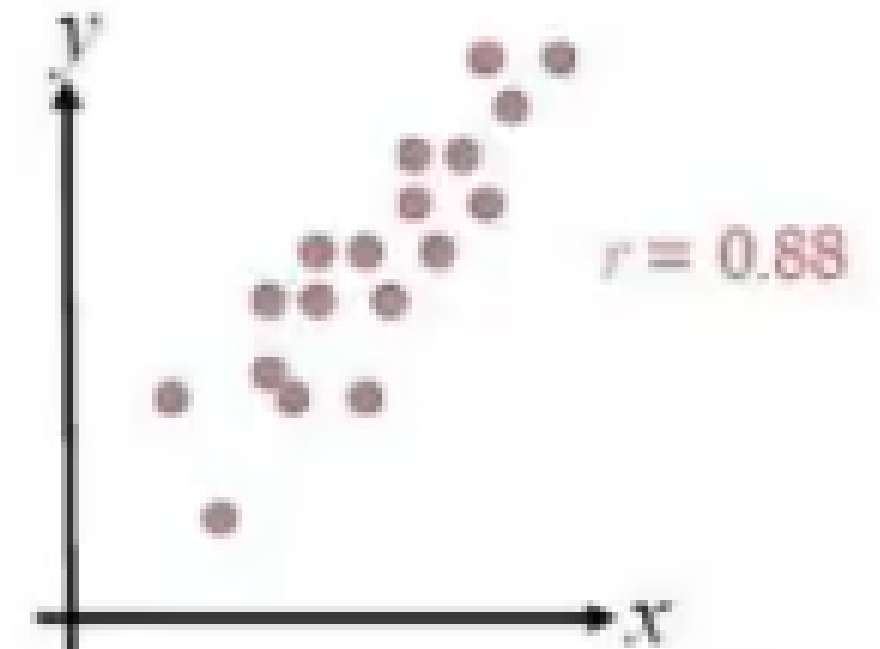
- The range of the correlation coefficient is **-1 to 1**.

CORRELATION COEFFICIENT

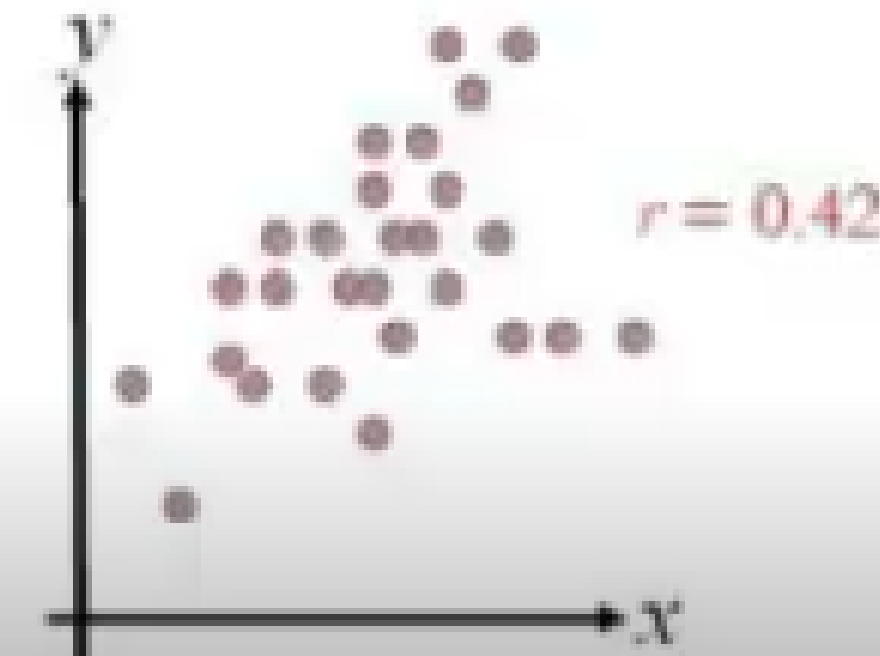
- $-1 \leq r \leq +1$
- If x and y have a strong positive linear correlation, r is close to 1.
- If x and y have a strong negative linear correlation, r is close to -1 .
- If there is no linear correlation or a weak linear correlation, r is close to 0.



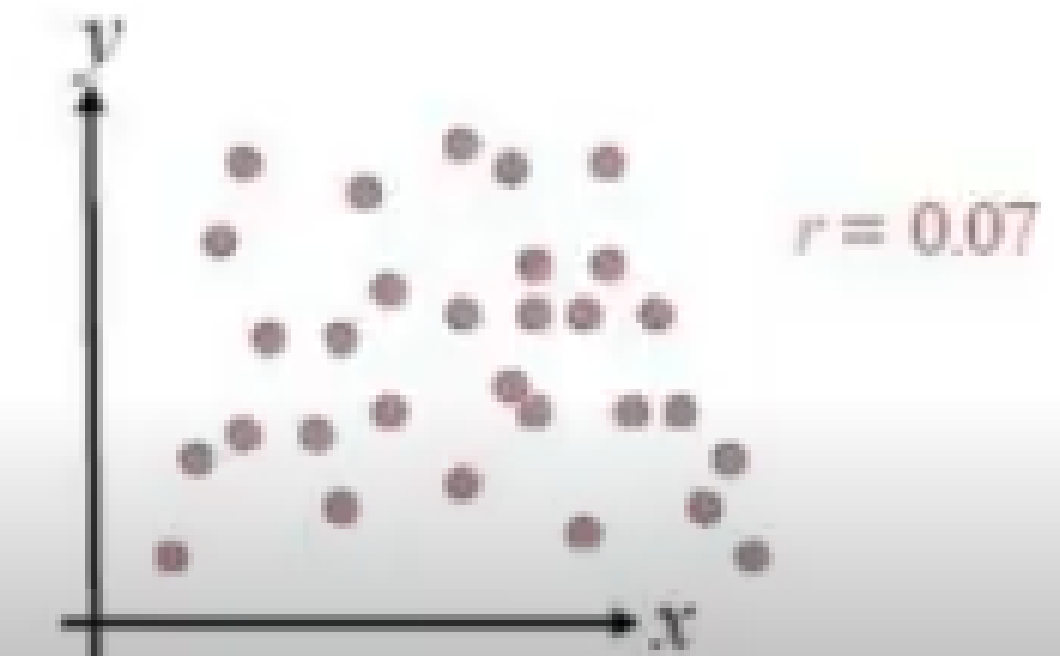
Strong negative correlation



Strong positive correlation



Weak positive correlation



Nonlinear Correlation

EXAMPLE 1

Calculate the correlation coefficient r for the following data.

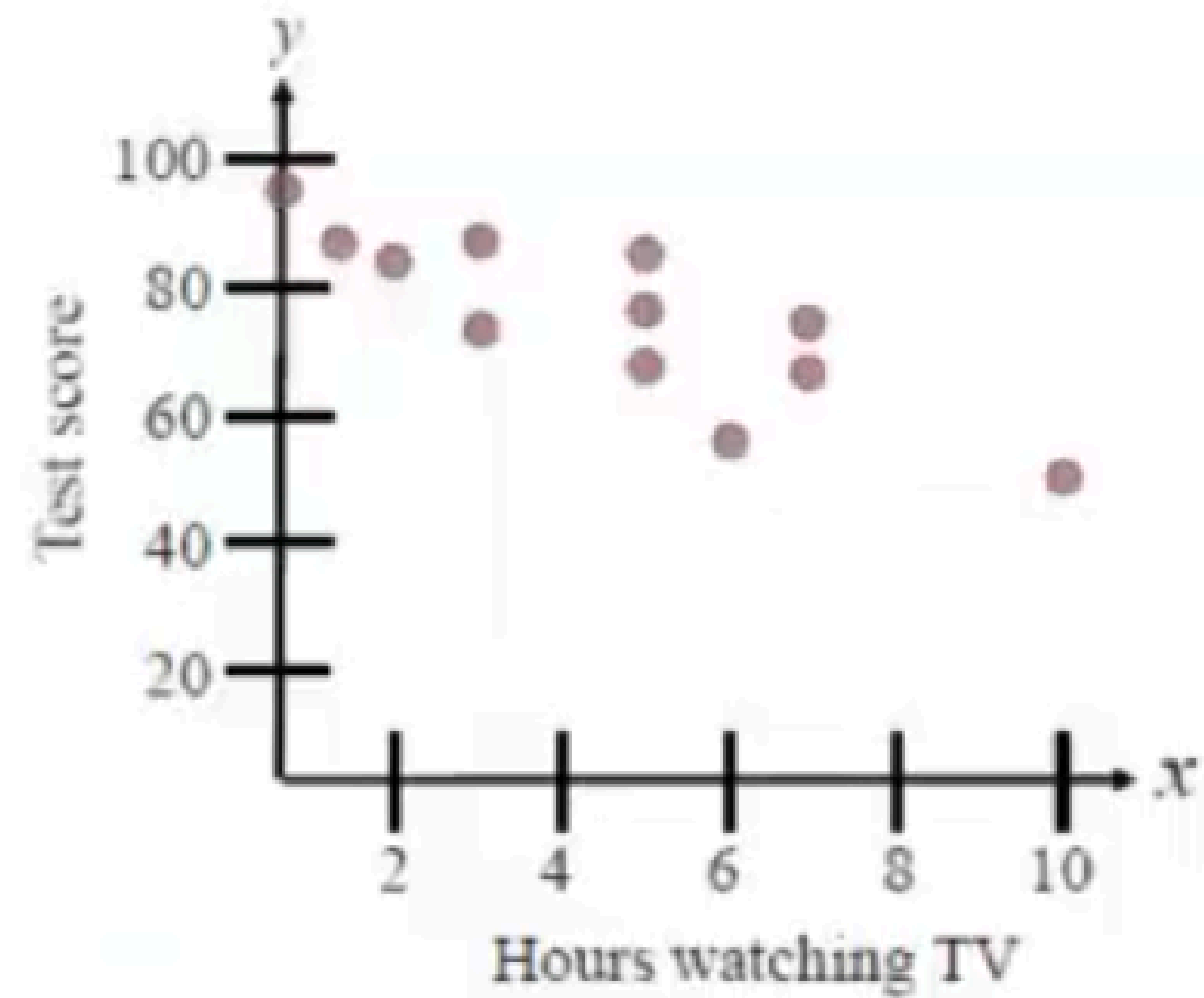
x	y	xy	x^2	y^2
1	-3	-3	1	9
2	-1	-2	4	1
3	0	0	9	0
4	1	4	16	1
5	2	10	25	4
$\Sigma x = 15$	$\Sigma y = -1$	$\Sigma xy = 9$	$\Sigma x^2 = 55$	$\Sigma y^2 = 15$

$$r = \frac{n\Sigma xy - (\Sigma x)(\Sigma y)}{\sqrt{n\Sigma x^2 - (\Sigma x)^2}\sqrt{n\Sigma y^2 - (\Sigma y)^2}} = \frac{5(9) - (15)(-1)}{\sqrt{5(55) - 15^2}\sqrt{5(15) - (-1)^2}} \approx 0.986$$

There is a strong positive linear correlation between x and y .

EXAMPLE 2

Hours, x	0	1	2	3	3	5	5	5	6	7	7	10
Test score, y	96	85	82	74	95	68	76	84	58	65	75	50



EXAMPLE 2

Hours, x	0	1	2	3	3	5	5	5	6	7	7	10	54
Test score, y	96	85	82	74	95	68	76	84	58	65	75	50	908
xy	0	85	164	222	285	340	380	420	348	455	525	500	3724
x^2	0	1	4	9	9	25	25	25	36	49	49	100	332
y^2	9216	7225	6724	5476	9025	4624	5776	7056	3364	4225	5625	2500	70836

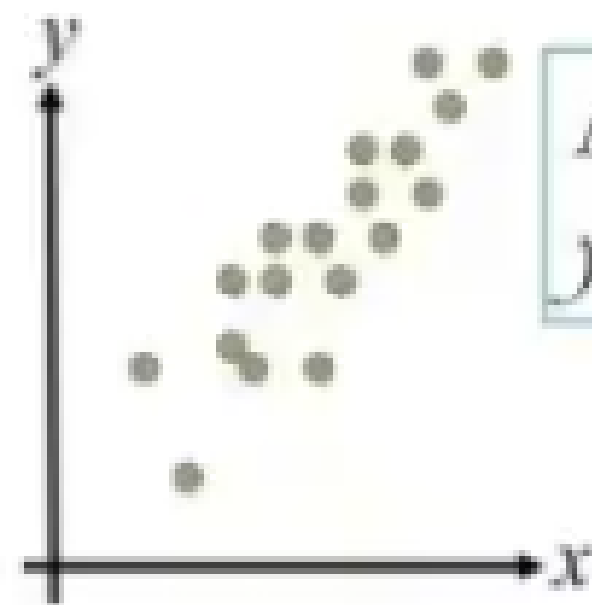
$$r = \frac{n\sum xy - (\sum x)(\sum y)}{\sqrt{n\sum x^2 - (\sum x)^2} \sqrt{n\sum y^2 - (\sum y)^2}} = \frac{12(3724) - (54)(908)}{\sqrt{12(332) - 54^2} \sqrt{12(70836) - (908)^2}} \approx -0.831$$

- There is a strong negative linear correlation.
- As the number of hours spent watching TV increases, the test scores tend to decrease.

COVARIANCE

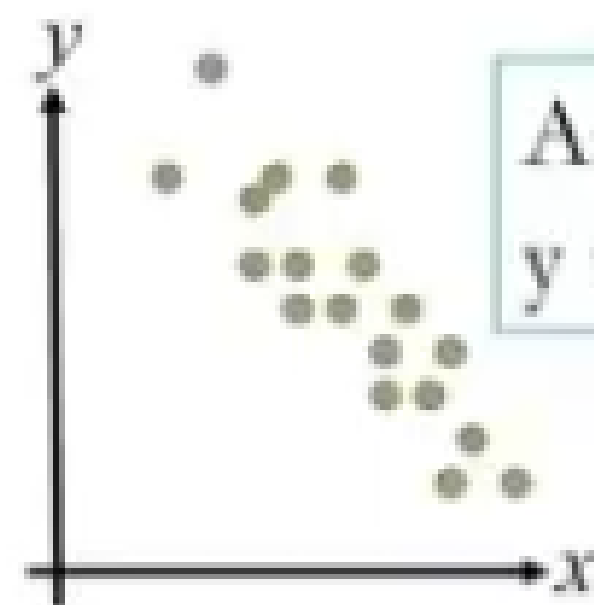
- **Covariance** is a measure of the relationship between two random variables.
- **Positive covariance:** Indicates that two variables tend to move in the same direction.
- **Negative covariance:** Reveals that two variables tend to move in inverse directions.

COVARIANCE



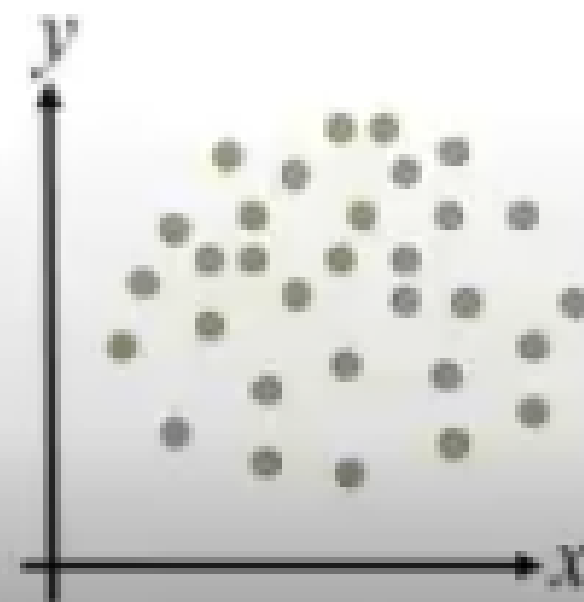
Positive Covariance

As x increases,
 y tends to increase.



Negative Covariance

As x increases,
 y tends to decrease.



Zero Covariance

COVARIANCE

$$\begin{aligned} Var &= \sigma_x^2 = \frac{\sum (X_i - \mu_x)^2}{N} \\ &= \frac{\sum (X_i - \mu_x) * (X_i - \mu_x)}{N} \end{aligned}$$

Where:

X_i - the values of the X-variable

Y_i - the values of the Y-variable

$\bar{X}$ - the mean (average) of the X-variable

$\bar{Y}$ - the mean (average) of the Y-variable

n - the number of data points

- Population covariance

$$Cov(X, Y) = \frac{\sum (X_i - \mu_x) * (Y_i - \mu_y)}{N}$$

- Sample covariance

$$Cov(X, Y) = \frac{\sum (X_i - \bar{X}) * (Y_i - \bar{Y})}{n - 1}$$